exoARM Folio

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Criteria 1: Design Brief

Context:

Statistics indicate that physical injuries sustained from work are a prevalent issue in Australia:

> 497,300 people had a work-related injury or illness (3.5% of people who worked at some time during the year). The most common cause of injury or illness was **'Lifting, pushing, pulling or bending'** (24%). 66% had time off as a result of the injury or illness. 31% received workers compensation for the injury or illness. (*Australian Bureau of Statistics*, 2023)

These issues can be alleviated through the implementation of assistive exoskeleton devices into workplaces. By creating an exoskeleton device which increases the force output through the entire range of motion of the joint, the muscular exertion required to execute lifting, pushing, or pulling motions would decrease. Since there is a correlation between physical exertion and rate of injury (Matos et al., 2020), a decrease in the required muscular exertion would decrease risk of workplace injuries. The exoARM system aims to increase the effective force output of the elbow joint, effectively decreasing the muscular work required to lift, push, or pull objects using that joint. This will decrease the risk of muscular strain and fatigue, especially for prolonged physical work. This is beneficial for both employers and employees, as not only will it increase the efficiency of physical tasks, by facilitating expedited completion, but it will also reduce the risk of workers being injured, and hence reduce the need for WHS compensation payments or leave.

This project will be focused on hinge joints, as they are some of the most load bearing and frequently used joints in the body.

> The hinge joints of the body include the elbow, knee, interphalangeal (IP) joints of the hand and foot and the tibiotalar joint of the ankle. (Gupton et al., 2023)

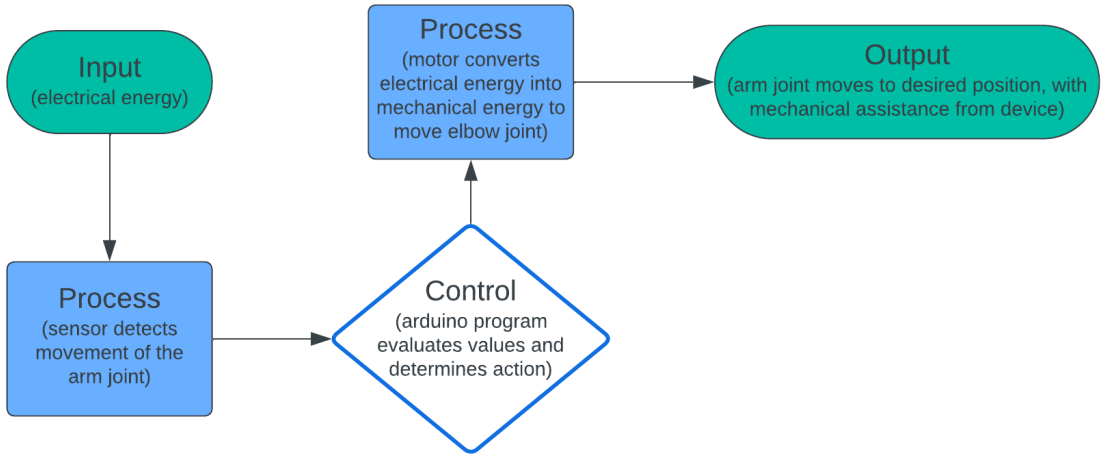
Because of their high frequency of use, hinge joints such as ankles, elbows, and knees are very commonly injured (*Avoiding Joint Injuries*, n.d.).

The exoARM device would be attached to the upper arm and forearm of the user, and also possibly supported by the back. It could use a sensor to detect the force that the user applies into moving the arm joint. The torque of this movement may effectively be amplified by the motor, which is controlled by an arduino program. The exoARM would also increase the strength of the elbow joint in static positions, such as when holding something out in front of the user.

References:

- https://www.ncbi.nlm.nih.gov/books/NBK518967
- https://www.abs.gov.au/statistics/labour/earnings-and-working-conditions/work-related-injuries/latest-release
- https://www.mdpi.com/1660-4601/17/11/4037
- https://www.stanfordchildrens.org/en/topic/default?id=avoiding-joint-injuries-1-2842]
- https://study.com/academy/lesson/mechanical-advantage-definition-formula.html

General IPO Diagram of basic features:



Glossary:

User: the term “User” is used throughout this report; it refers to an individual who is wearing and operating the functional exoARM device.

Device: the term “Device” is used throughout this report; it refers to the complete exoARM device.

Implementation Constraints:

The project has several constraints from various areas, including:

- The funding of the project received from Viewbank College is limited to $100, however this can be increased through external funding.
- The time to complete the project is limited to the duration of the VCE Systems Engineering 1 & 2 course. This is approximately 20 weeks of development.
- The device must be lightweight enough to avoid inducing muscular strain or mechanical strain on joints due to its weight.
- The device must be compact enough to not inhibit freedom of arm movement when working.
- The device must increase the effective mechanical lifting force of the elbow joint when lifting, pulling, or pushing objects during work.
- The device must have tolerance range and speed limits to avoid straining muscles or harming joints by pushing them beyond reasonable ranges of motion or speeds.
    - Must incorporate safety mechanisms to prevent over-assistance, which could potentially cause injury or strain to the user.
    - Should include fail-safe features to immediately disengage the assistance in case of malfunction or user discomfort.
- The device must be able to detect both the position of the elbow joint as well as the acceleration.
- The device must be able to be easily put on or taken off. It also must be able to easily be operated and disengaged.
- All components in the device must be easily obtainable and easily hot swappable in case of failure. This means that wires should not be soldered in place, and should have standard connectors.

- The device should be designed with ergonomic features to ensure user comfort during prolonged use. It must also allow supination and pronation of the forearm while still being able to provide assistance.
- Padding or cushioning should be used throughout the device to minimise pressure points and discomfort on the user's arm.
- The device should allow for adjustability to accommodate different users with varying arm sizes and strength levels.
- The device should have adjustable straps or fasteners to secure it comfortably to the user's arm.

Considerations:

The following considerations can be taken into account to improve the efficiency, quality, and effectiveness of the device.

- The ability to increase battery life of the device by using renewable energy sources like thermoelectric generators or solar panels.
    - This would improve the user experience as they would not need to charge the device as frequently, and it would also allow the device to have a smaller battery, which would decrease weight.
- A user-friendly interface, such as an online interface or physical buttons and dials
    - This could help improve the accessibility of the device, allowing its settings, like torque, speed and mode, to be toggled without requiring knowledge of programming.
- Sensors which monitor heart rate, and other similar information.
    - This could help ensure that the user is safe, by monitoring if the different values are within their required ranges.

Evaluation Criteria:

Criteria	Reason for assessment	Method of assessment
Was the project completed within the $100 budget?	This is measured to ensure that the device remains below the budget limit set by the school. This will also ensure that the device is affordable and accessible to all Australians in need of the device.	Keep record of all expenses in the project and compare it to the initial budget.
Was the project completed within the time constraint of 20 weeks?	This is assessed as the client requires the device to be submitted before a specified date.	Record the start date of the project and submission date to calculate the time taken to complete the project
exoARM criteria:		
Is the device must be lightweight enough to avoid inducing muscular strain or mechanical strain on joints?	This is measured to ensure the safety and comfort of users when using the exoARM device for prolonged periods of time. If the device is heavy, it could counterproductively lead to an increased rate of fatigue, discomfort, or strain over time.	Assessing the device's weight against established ergonomic standards and user feedback after prolonged trials (10 minutes)
Does the device allow freedom of arm movement?	This is measured to ensure that the exoARM device facilitates increased productivity in workplace environments, maintaining manoeuvrability. If the exoARM restricts movement, it would decrease performance in the workplace and also cause safety hazards.	Measuring the range of motion of the elbow joint permitted by the device (in degrees), and by conducting trials performing real world tasks while wearing the device.
How much does the device increase the effective pulling and pushing force of the arm?	This is measured to evaluate the functional effectiveness of the exoARM device for lifting, pushing, or pulling tasks. The extent to which the exoARM device enhances the user's capacity to exert force also gives insight into which types of workplace applications it is appropriate for.	Comparative assessments lifting dumbbells in a bicep curl motion with and without the device augmentation, and recording the highest weight that can be lifted with and without the device.

Are the tolerance range and speed limits effective, avoiding excessive ranges of motion or speeds.	This is measured to ensure that the exoARM device operates within the natural range of motion and speeds of the human arm. Excessive speeds or ranges of motion could counterproductively lead to an increased rate of fatigue, discomfort, or strain over time.	Having users perform real-world tasks and provide feedback on any discomfort, fatigue, or strain.
Is the device simple to operate, adjust, and disengage?	This is assessed to ensure that the exo RAM is usable and accepted by users in real-world settings. Complex or time-consuming attachment procedures could deter users from utilising the device consistently.	Assessing the simplicity and efficiency of the device's fastening mechanisms through user trials. Time taken to put on and activate the device should be measured.
Is the device easily repairable?	This is assessed to ensure that the exoARM device is cost-effective and reliable for long term use.	Assessing the device's repairability through disassembly trials and component replacement simulations. All components in the device should ideally be easily obtainable and easily hot swappable in case of failure. This means that wires should not be soldered in place, and should have standard connectors.
Is the latency of the device low enough to be able to use it effectively?	This is assessed to ensure that the device is user friendly and both intuitive and safe to operate.	Use a video recording to measure the difference in time between the input (e.g. button press) and output (e.g. movement of arm)
Are the structural components of the system strong enough to bear heavy loads?	This is assessed to ensure the longevity and safety of the device. If the structural components fail under load, it could lead to injury. Also, the device must be able to maintain structural integrity to meet other criteria regarding the force exertion.	After applying mechanical stress to the system from usage, visually observe if there are any visible deformities or cracks.

Factors influencing Design, Planning, Production, and Use:

Factors influencing Design:

- Safety of Users (Workers and Employers):
    - The exoARM must be designed to ensure the safety of both workers and employers. This involves preventing overexertion or misuse that could lead to injuries, aligning with ergonomic standards.
- Functionality:
    - The device must effectively increase the force output of the elbow joint without compromising other functions. It should detect the user's movements and amplify them accurately.
- Aesthetics:
    - While not the primary concern, the device should have an appealing design, encouraging user adoption and reducing any psychological barriers associated with wearing the device.
- Cost of Components:
    - To stay within the allocated budget, components should be chosen based on their cost-effectiveness without sacrificing quality. The design should meet the $100 budget.
- Materials and Components Selection:
    - The materials used should be durable, lightweight, and preferably sourced sustainably. Components should be chosen for their longevity and minimal environmental impact.

Factors influencing Planning:

- Time Available for Production:
    - The project must be planned within the 20-week timeframe, including all stages from design to testing. The schedule must be meticulously followed to meet deadlines.
- Availability of Materials and Components:
    - Sourcing materials quickly without compromising quality is crucial. If certain components are delayed, alternatives must be considered to maintain the project timeline.
- Availability of Equipment and Learning Curves:
    - Planning should account for the availability of necessary equipment and the time needed to train on new tools. This includes allocating time for learning and ensuring equipment access.
- Risk Management:
    - The risks associated with using certain tools and processes must be minimised. This includes conducting risk assessments and implementing safety measures during production.

Factors influencing Production:

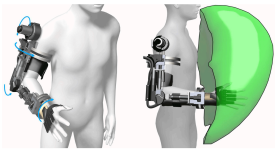
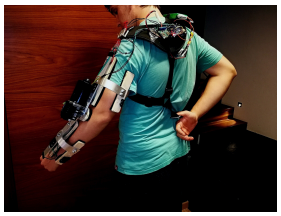
- Safety:
    - Production must adhere to safety standards to ensure the safety of the developers and any participants during testing phases.
- Knowledge of Process and Use of Equipment:
    - The team's familiarity with the production processes and equipment will influence the quality and speed of production. Training or consulting with experts may be necessary.
- Quality Control:
    - Maintaining high-quality standards throughout production is essential. Each component and the final product must be tested for functionality and durability.
- Step-by-Step Production Plan:
    - A detailed, step-by-step plan should guide the production process, ensuring all tasks are completed efficiently and within the set timeline.

Factors influencing Use:

- Safety for Users:
    - The exoARM must be safe for prolonged use. It should avoid sharp edges, rough surfaces, and exposed electronics, ensuring comfort and safety during operation.
- Outdoor Weather Conditions:
    - If the device is to be used outdoors, it must be weatherproof and durable, protecting against elements like rain, dust, and temperature variations.
- Environmental Sustainability:
    - The exoARM should have a minimal environmental impact, using sustainable materials and components that consume minimal power. The design should facilitate easy repair to extend the device's lifespan.
- Cost of Final Product:
    - The final product should be affordable for users, both in terms of initial purchase and ongoing maintenance. Features that reduce the need for frequent interventions (e.g., battery replacements) are essential.

Criteria 2: Researching, Designing, and Modelling and Design Options

Research of similar products

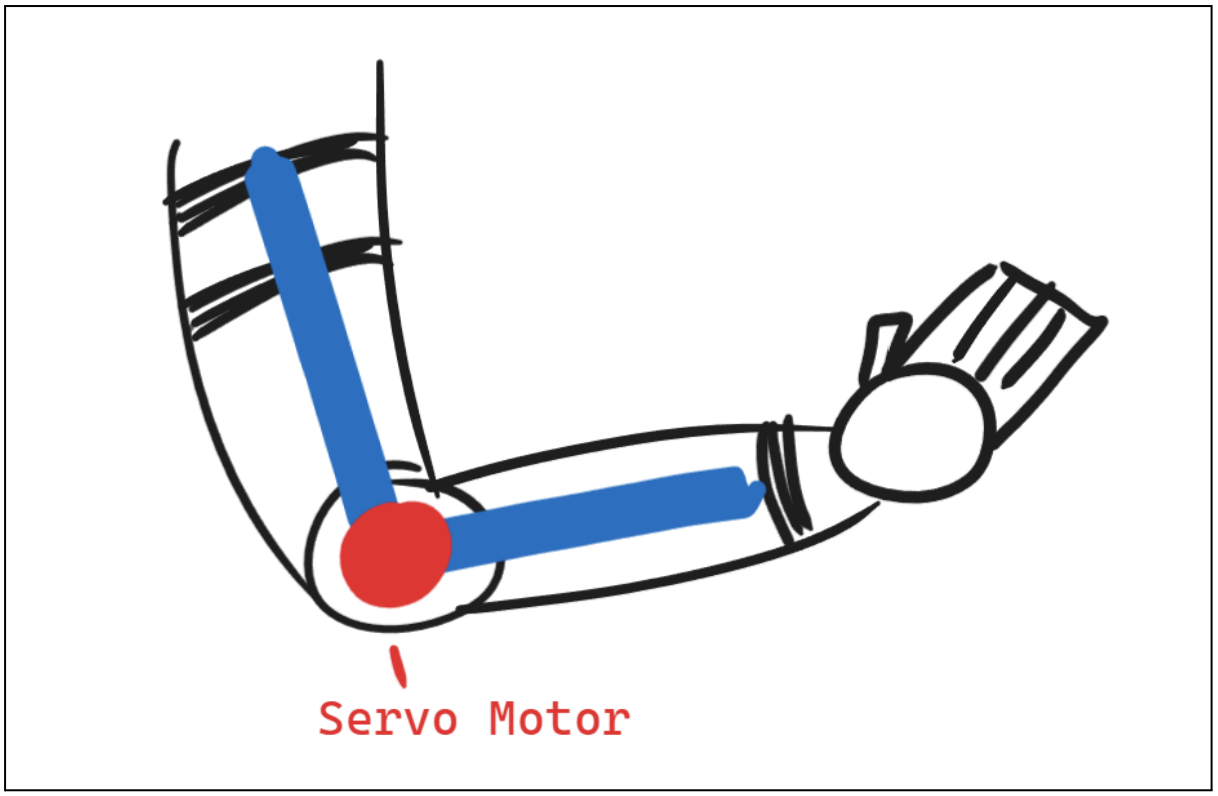
Product:	Potential similarities:	Potential differences:	Summary:
	- Attached at the upper arm and forearm area - Provides force amplification for the elbow joint	- Free swivelling wrist movement - Uses motors for actuation	This product is far more mechanically and technically complex than my project, which would lead it to have a much higher cost and therefore may not be accessible to people in need of the technology. My device will keep a similar function to this project, but with far more simplicity.
	- Attached at the upper arm and forearm area as well as the back - Supports elbow joint movement	- Uses motors for actuation	This product presents an affordable approach to solving the issue of my project, however it is based on a windscreen wiper motor. This means it sacrifices force output to achieve such a low cost. The way that the device is supported by the upper back is noteworthy and could be investigated.
	- Attached at the upper arm and forearm area - Provides force amplification for the elbow joint	- 3rd class lever	This product's use of mechanical muscle actuation and torso support provides an alternative approach to force amplification, which was considered for the device, but ultimately ruled out

			due to concerns with force output and range of motion.
	- Attached at the upper arm and forearm area - Provides force amplification for the elbow joint - Uses velcro for attachment to arm	- Uses motors for actuation - Uses a pulley system	This product's shoulder and torso harness offer an alternative approach to distributing the weight and force, which could enhance user comfort and reduce strain, offering valuable design considerations for the device.

Design

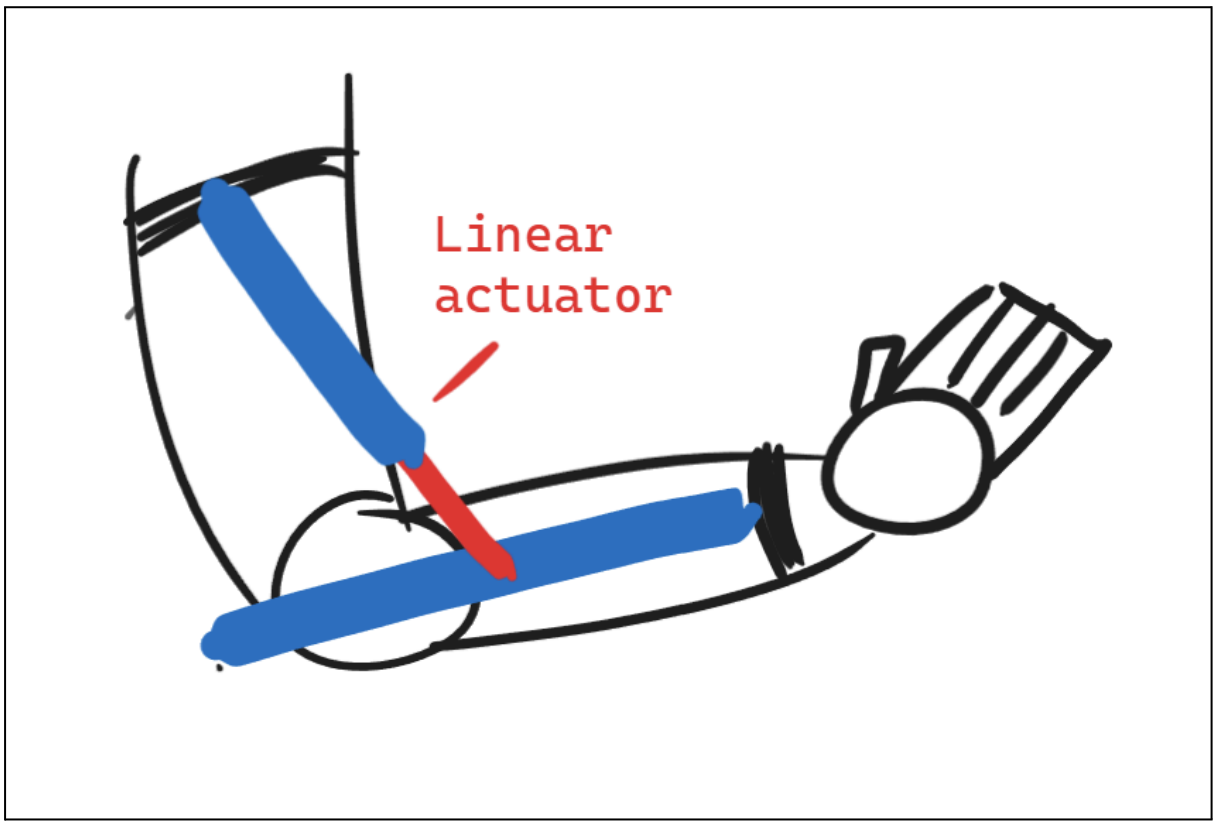
Motor & Lever Options

Servo Motor (Class 3 Lever)



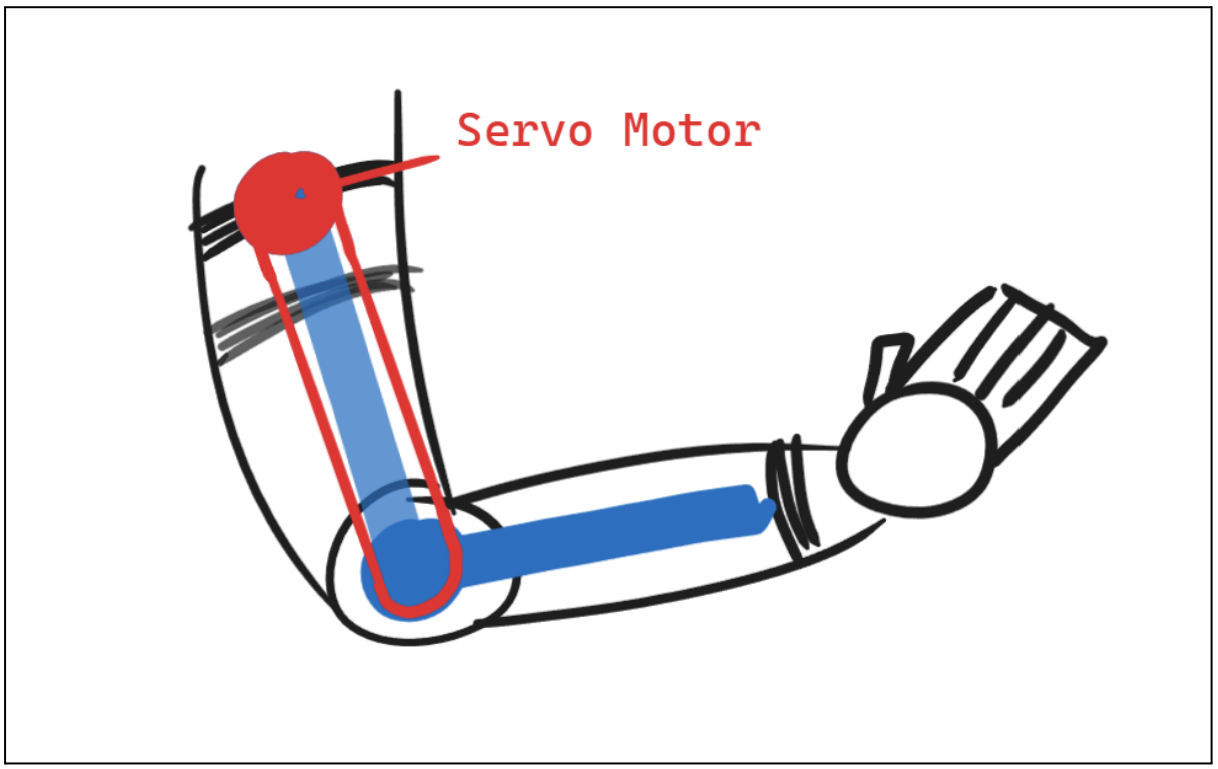
- Evaluation: The servo motor was immediately ruled out after simple torque calculations revealed that it did not provide sufficient torque to drive the arm effectively.
- Conclusion: Insufficient torque output made this option non-viable for the exAM.

Linear Actuator (Class 3 Lever)



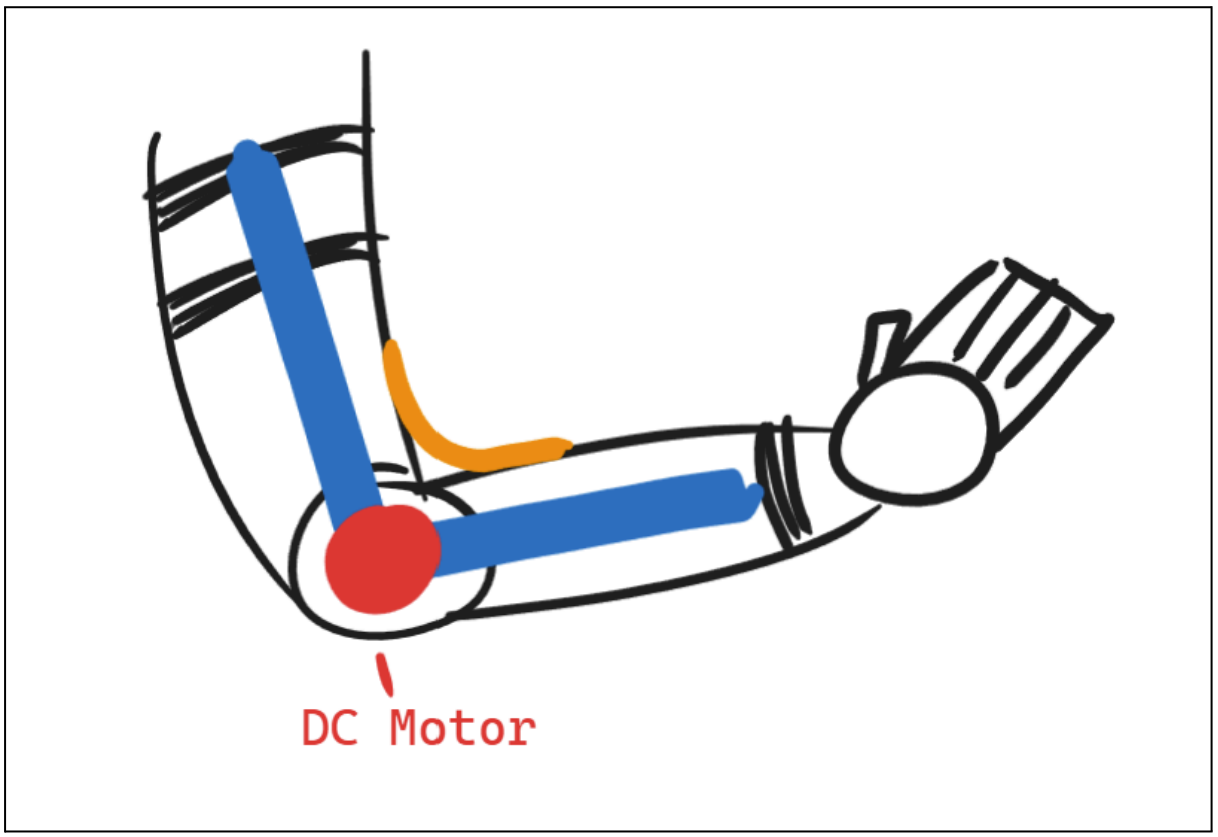
- Evaluation: While the linear actuator offered a simpler mechanism and substantial force production, it greatly restricted the user's freedom of movement.
- Advantages: Simpler mechanism, high force production.
- Disadvantages: Severely limited freedom of movement, making it impractical for the exAM.
- Conclusion: Restricted movement disqualified this option despite its force advantages.

Pulley with Servo Motor (Class 3 Lever)



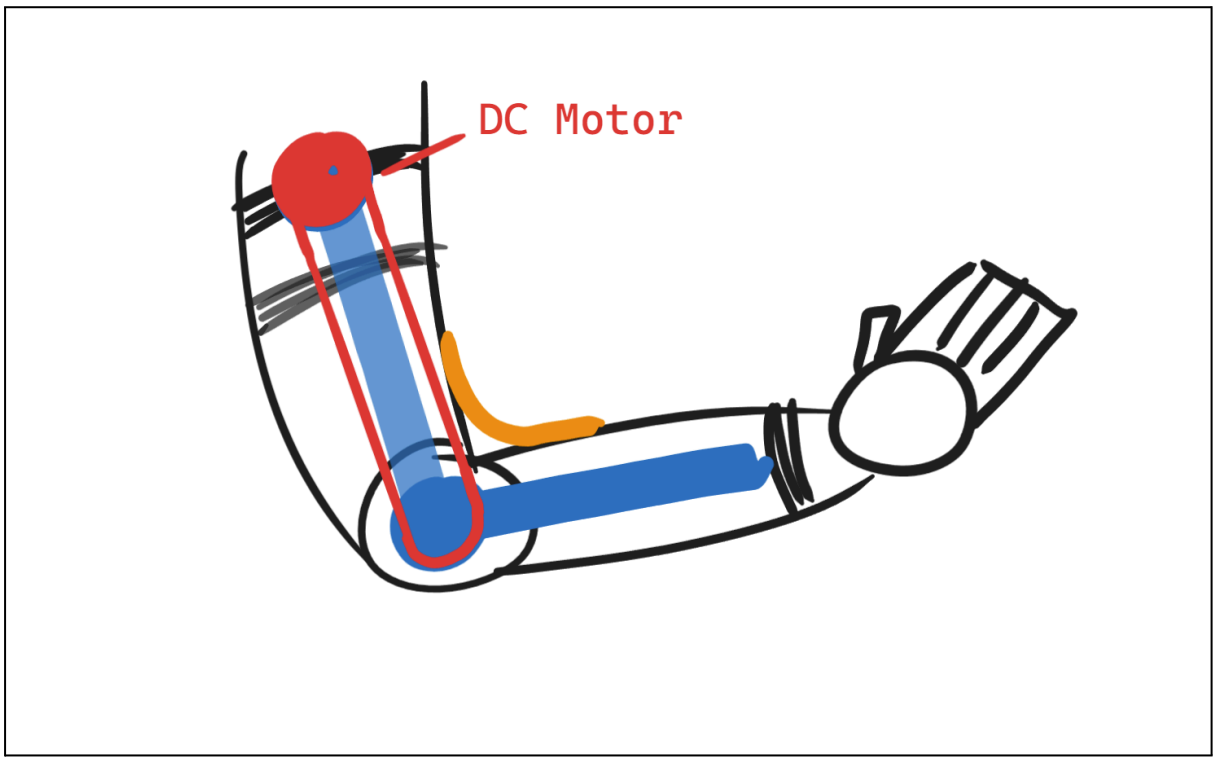
- Evaluation: Similar to the direct servo motor configuration, the torque was insufficient even when paired with a pulley system.
- Conclusion: Inadequate torque output remained an issue, ruling out this combination.

DC Motor + Flex Sensor (Class 3 Lever)



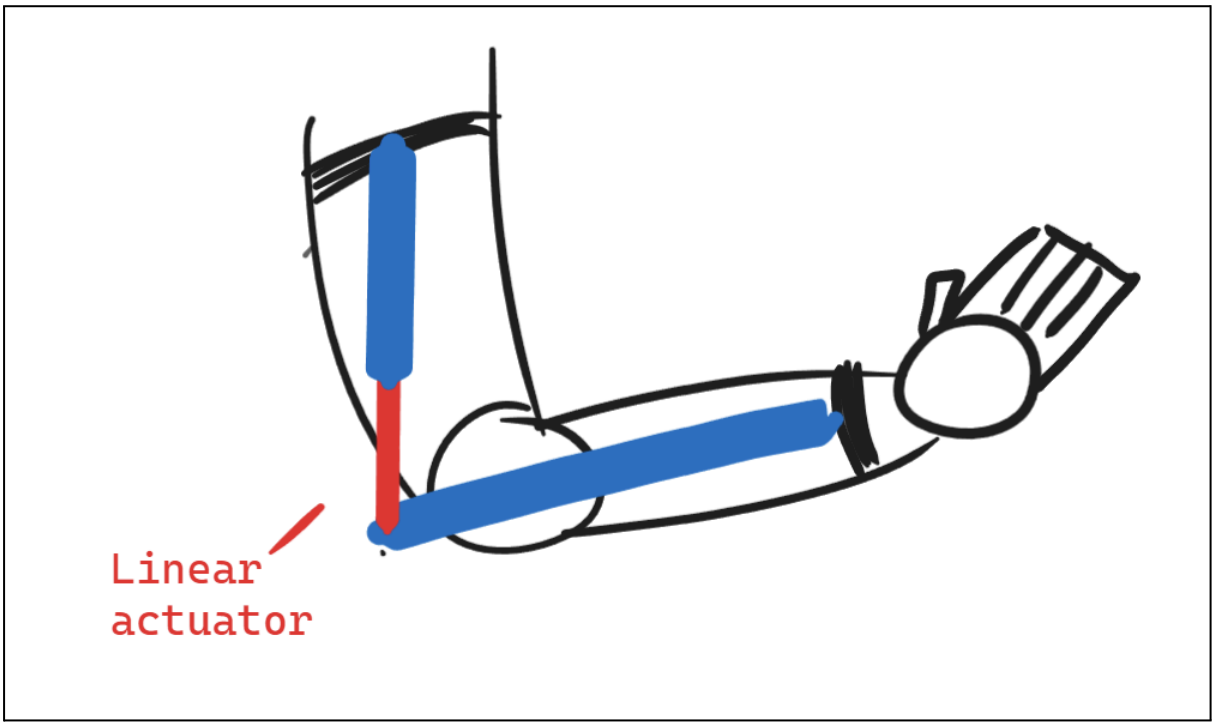
- Evaluation: This combination was a strong candidate due to its potential for controlled movement. However, the required torque of approximately 250kgF per cm was far beyond the budget constraints and significantly increased the device's weight.
- Advantages: High control precision, effective force augmentation.
- Disadvantages: Excessive torque requirements, budget limitations, increased weight.
- Conclusion: High torque and budget constraints made this option impractical.

Pulley with DC Motor + Flex Sensor (Class 3 Lever)



- Evaluation: The addition of a pulley aimed to reduce the required motor force by half. However, this configuration still faced issues with force production and added mechanical complexity, increasing the risk of faults.
- Advantages: Potential reduction in motor force requirements.
- Disadvantages: Continued issues with sufficient force production, increased mechanical complexity.
- Conclusion: The complexity and insufficient force production rendered this option unsuitable.

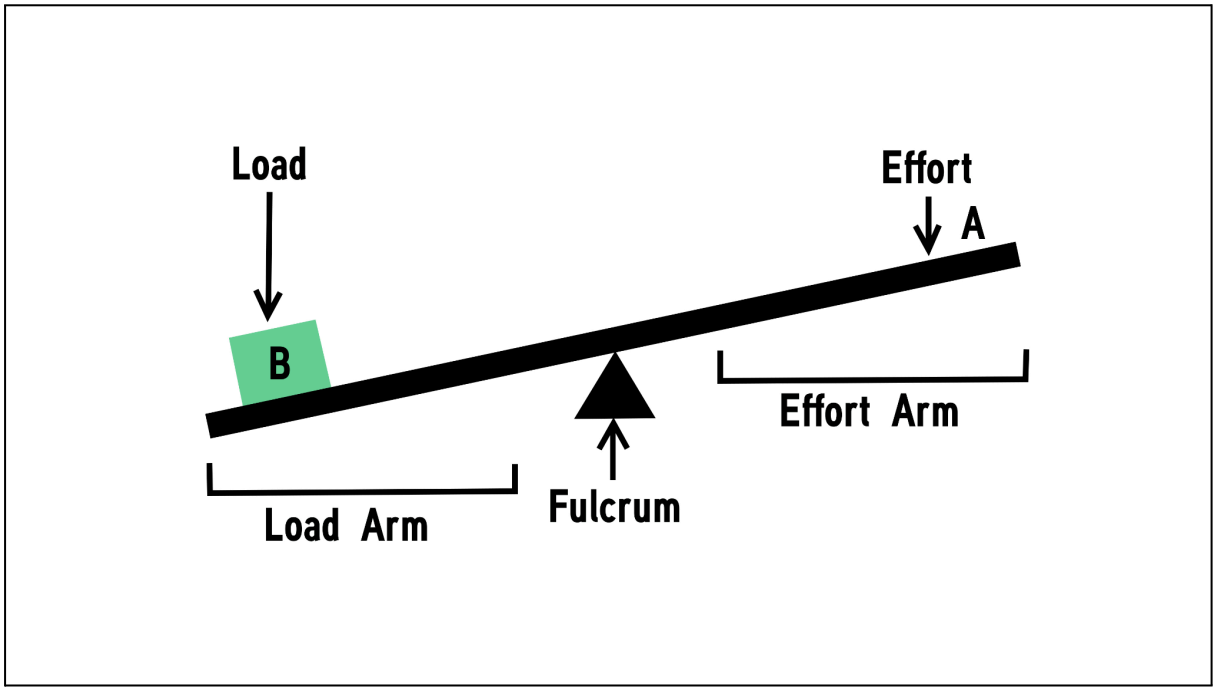
Linear Actuator (Class 1 Lever)



- Evaluation: Ultimately, the linear actuator in a class 1 lever configuration was selected. This setup provided the highest amount of force within the budget and maintained simplicity in the mechanism, which facilitated easier calculations and a straightforward design.
- Advantages: High force output, budget-friendly, simple mechanism, easier calculations and design implementation.
- Disadvantages: Potentially less natural movement compared to more complex systems, but this was a trade-off for practicality and budget adherence.
- Conclusion: This option was selected for its balance of high force output, simplicity, and adherence to budget constraints, making it the most feasible and effective choice for the exAM.

Lever Options

The device's main mechanical feature is a first class lever. It has a central fulcrum, and a load on either side of the fulcrum, as shown in the diagram below.



The linear actuator will exert a maximum downwards force of 1500N on the effort arm. To determine the force exerted on the load arm, mechanical advantage must be calculated. Mechanical advantage is a measure of the ratio of output force to input force in a system.

$$Mechanical\ Advantage\ (MA) = \frac{Effort\ arm\ length}{Load\ arm\ length} = \frac{Output\ force}{Input\ force}$$

Since the maximum input force of the system is 1500N, the formula below can be derived algebraically.

$$F_{output} = 1500 \times \frac{Effort\ arm\ length}{Load\ arm\ length}$$

The average forearm length is approximately 280mm. Therefore, within the calculations, I decided to set the load arm length at a fixed length of 280mm which should suit most users, leading to the formula below, where effort arm length is denoted by L_e:

$$F_{output} = 1500 \times \frac{L_e}{280}$$

Note: For users with shorter arms, the maximum force output of the device would be slightly increased, and for users with longer arms, the maximum force output of the device would be slightly decreased.

The mechanical advantage of the device also has an effect on the displacement of the load. The linear actuator selected has a 100mm stroke, meaning that the input displacement is 100mm. The output displacement is calculated by the following formula:

$$s_{output} = 100 \times \frac{280}{L_e}$$

Additionally, using the cosine rule, the following formula can be derived to determine the range of motion of the device in degrees:

$$\theta_{range\ of\ motion} = \cos^{-1}\left(\frac{2\times {L_e}^2 - 100^2}{2\times {L_e}^2}\right)$$

Option 1: 28mm effort arm

Criteria	**Value**	**Comments**
Mechanical advantage:	1:10	
Maximum force:	150N	
Maximum weight:	15kg	This is insufficient for heavy lifting applications.
Maximum range of motion:	Unlimited	The full 100mm displacement (and therefore the maximum work) of the linear actuator is not able to be used.

Option 2: 70mm effort arm

Criteria	**Value**	**Comments**
Mechanical advantage:	1:4	
Maximum force:	375N	
Maximum weight:	38kg	This is sufficient for most heavy lifting applications.
Maximum range of motion:	91°	This range of motion may be excessive for practical use.

Option 3: 100mm effort arm

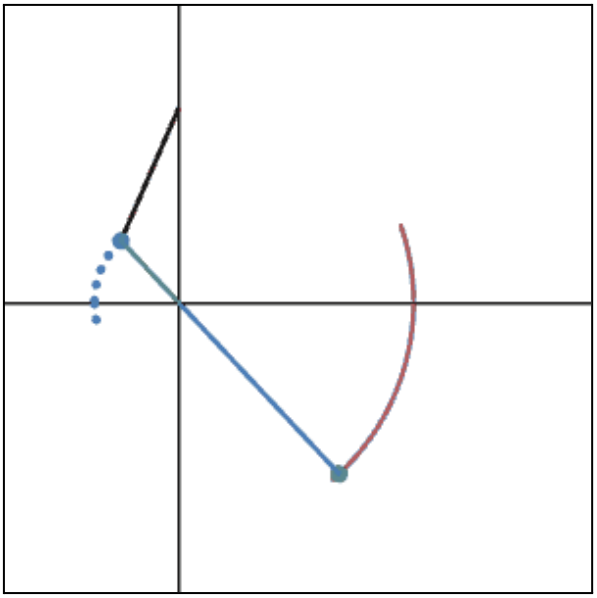
Criteria	**Value**	**Comments**

Mechanical advantage:	5:14	
Maximum force:	535N	
Maximum weight:	55kg	This is more than sufficient for most heavy lifting applications.
Maximum range of motion:	60°	The decreased range of motion decreases the risk of injury to the user.

Option 4: 140mm effort arm

Criteria	**Value**	**Comments**
Mechanical advantage:	1:2	
Maximum force:	750N	
Maximum weight:	76kg	There are no safe applications for lifting this much weight as it may put other parts of the body at risk.
Maximum range of motion:	42°	This range of motion is insufficient for practical use.

Option 3 was selected. The maximum lifting force of 535N allows any heavy lifting to be completed easily. Additionally, the 60° range of motion increases safety by preventing hyperextension of the elbow joint.



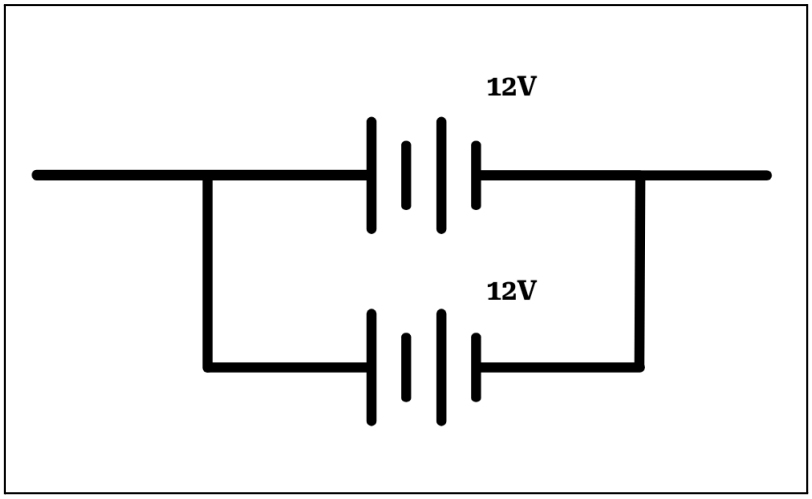
The above diagram visualises the simple lever mechanism of the device (blue and green lines), going through its full 60° range of motion. The linear actuator (black) is shown extending from 0mm to 100mm, its full displacement range.

Battery Calculations and Circuit Diagram

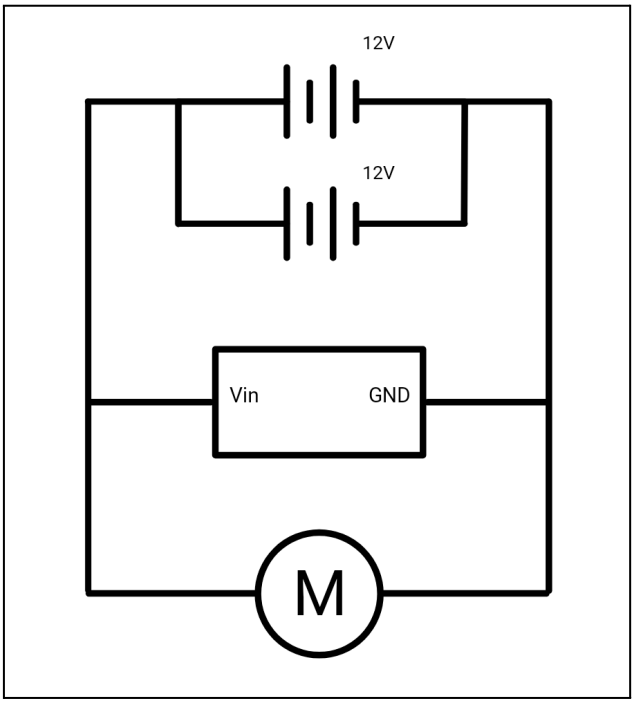
The linear actuator being used is a 12V DC motor, and draws a maximum current of 3A.

To achieve this current and voltage, it was decided to use two 12V 3000mAh Lithium Ion batteries. Each battery has a maximum current output of 2A which is slightly less than the specifications of the linear actuator.

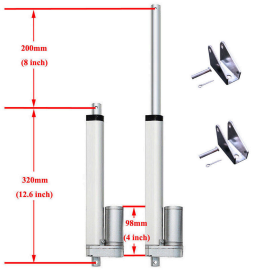
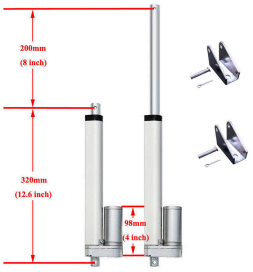
These batteries would be wired in a parallel configuration as shown below, increasing the maximum current output.



The circuit diagram below shows the configuration of the linear actuator, arduino, and batteries. Other electrical components are excluded for simplicity.



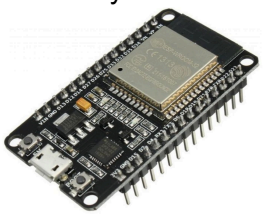
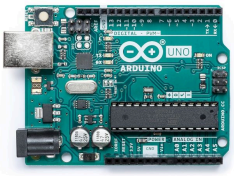
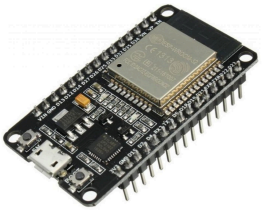
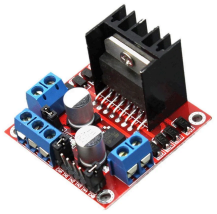
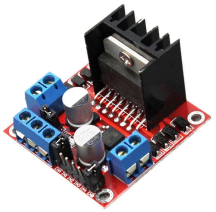
Research of Mechanical Components

Component:	Options:	Pros & Cons:	Decision and Justification:
Linear actuator 	Option 1: Heavy Duty 12V Linear Actuator 100mm 1500N (from eBay)  Option 2:DC 12V Linear Actuator, Stroke 100mm 12V Load Capacity 1000N (from Amazon)	Option 1: - Pros: Higher force and far cheaper - Cons: Longer shipping times Option 2: - Pros: Faster shipping times and better returns policy - Cons: Far more expensive	**Option 1** was selected as it was $51.99, compared to Option 2 which was $72.44. Additionally, the increased force output allowed for an increase in the amount of load which can be held with the device.
Load bearing axle	Option 1: M10 x 75mm Zinc Plated Hex Head Bolts and Nuts Option 2: M10 x 25mm Zinc Plated Hex Head Bolts and Nuts	Option 1: - Pros: Partially threaded; reducing friction. Longer, allowing more flexibility for thickness of parts. - Cons: Partial threading section is very long, meaning that the hex nut will not be securely fastened. Option 2: - Pros: Shorter; will not need to be	**Option 1** was selected. The longer bolt provided more flexibility in the thickness of the structural parts. Additionally, the partial threading gave the bolt a smooth, low friction surface to serve as the load bearing axle. However, the bolt will need to be cut shorter, removing the threaded section. The hex nut will need to be welded onto the smooth section of the bolt.

		shortened. Fully threaded, allowing hex nut to be used to securely fasten the mechanism. - Cons: Fully threaded, increasing friction and risk of wear from use. Shorter; may not be long enough to go through the entire mechanism.	
Hex nuts and bolts for brackets 	Option 1: Pinnacle M8 x 35mm Zinc Plated Hex Head Bolts And Nuts - 16 Pack Option 2: M8 x 20mm Zinc Plated Hex Head Bolts And Nuts - 18 Pack 	Option 1: - Pros: Longer bolt to allow for securing thicker components - Cons: Higher unit price Option 2: - Pros: Lower unit price - Cons: Shorter bolt	**Option 1** was selected due to having more flexibility for the possible addition of thicker parts. If necessary, the bolts can be cut to be shorter using available equipment.
Grease 	Option 1: White lithium grease Option 2: Silicone lubricant	Option 1: - Pros: Functions better with metals - Cons: N/A Option 2: - Pros: Multi-surface use - Cons: Does not work as well with metals	**Option 1** was selected as a better option due to it being designed for use with metals.

	WD-40 SPECIALIST SILICONE QUICK-DRYING		

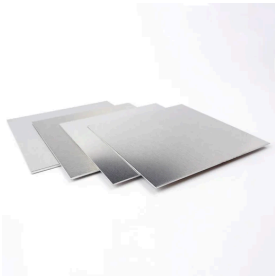
Research of Electrical Components

Component:	Options:	Pros & Cons:	Decision and Justification:
Control system 	Option 1: Arduino Uno R3  Option 2:ESP32 Wireless WiFi + Bluetooth Development Board 	Option 1: - Pros: Easily accessible from school technology supplies, supports high voltage power supply, more adjustability - Cons: larger Option 2: - Pros: Allows for wifi and bluetooth connection, allowing for the possibility of controlling device via an app - Cons:harder to obtain, longer shipping times	**Option 1** was selected for the prototype, as it is easily obtainable and has hot swappable connections to allow for easy adjustments.
Motor controller 	Option 1:DC motor controller bundled with linear actuator (Option 1) Option 2:L298N motor controller 	Option 1: - Pros: Designed to be used with the linear actuator, less features so simpler to operate - Cons: Less documentation, less support for ESP32 and arduino Option 2: - Pros: Large amounts of documentation and support for ESP32 and arduino. Smaller and more versatile. Available from school technology resources. - Cons:More	**Option 2** was selected due to its availability in school technology supplies, abundance of documentation, support for arduino and ESP32, and its versatility

		complicated to operate	
12V battery pack 	Option 1:12V Rechargeable Li-ion Battery Portable Battery Pack with AU Plug Charger (eBay) Option 2: 	Option 1: - Pros: Provides 12V which is the required voltage for the linear actuator. Rechargeable easily using the wall adapter included. Provides required current with two batteries. - Cons: Risk of overheating Option 2: - Pros: Increased safety and widely available. Easier to replace in case of failure. - Cons: Only provides 9V. Does not have a large capacity. Does not allow for sufficient current draw.	**Option 1** was selected due to its high voltage and current, simplicity, lightweight and small profile, and its ability to be recharged. Additionally, it has switches built-in to the casing of the device
Controller switch 	Option 1: SPDT Centre Off Rocker Switch Option 2: EEEkit 2 Pack 6 PIN DPDT 20 Amp Momentary Rocker Switch	Option 1: - Pros: Low unit price and easily available with no shipping time. - Cons: N/A Option 2: - Pros: Higher voltage and current specifications. DPDT rather than SPST - Cons: Higher unit price, longer shipping times.	**Option 1** was selected due to its lower unit price and high availability within Australia (from JayCar). Option 2 had some improved features, but they were ultimately not required as the switch will be connected to an arduino rather than to a battery directly.

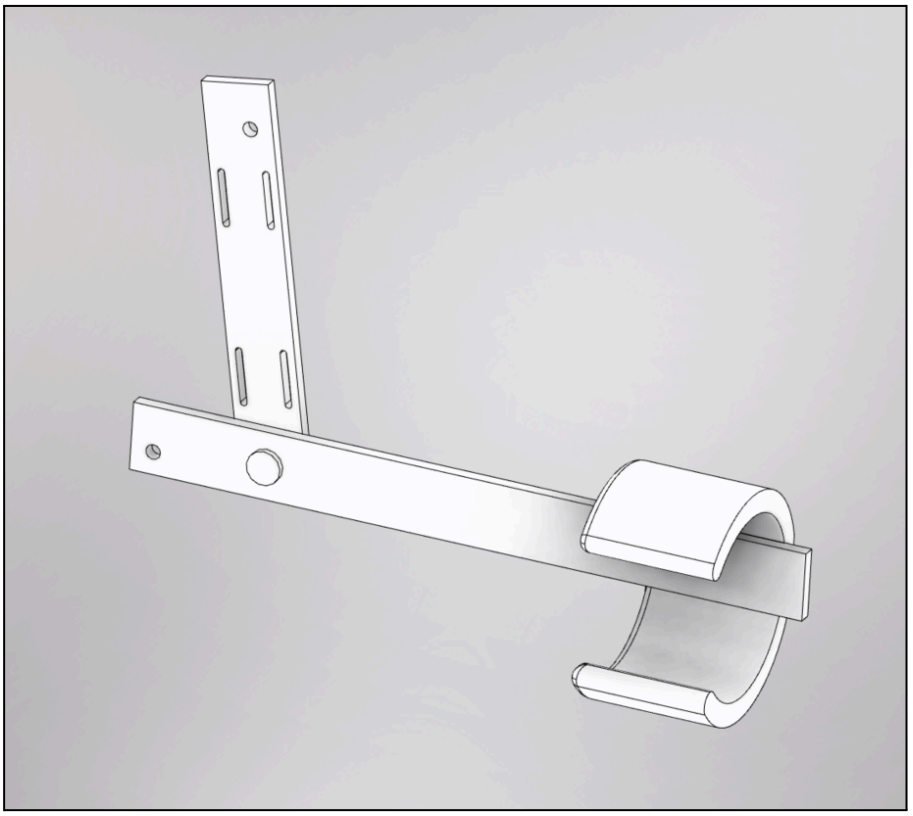
Wires 	Option 1: Dupont connector kit + crimping tool + wires Option 2: Jumper wires, assorted lengths and colours 	Option 1: - Pros: Can be cut to any length - Cons: Far more expensive, less secure, more time consuming Option 2: - Pros: Cheap and simple - Cons: Fixed length	**Option 2** was selected to reduce time consumption and cost.
Kill switch (Normally closed push button) 	Option 1: SPST Option 2: switch with LED 	Option 1: - Pros: Simpler - Cons: Lower maximum current Option 2: - Pros: Visual indication of current flow, higher maximum current - Cons: Higher unit price	**Option 2** was selected due to its functionality to visually indicate if the device is running or not, as well as its higher maximum current.

Research of Materials

Component:	**Options:**	**Pros & Cons:**	**Decision and Justification:**
Metal parts	Option 1: Aluminium sheet metal 8mm  Option 2: Galvanised square steel 20mm 	Option 1: - Pros: Can be precisely cut using water-jet cutting technology. Is lightweight. Readily available from school metalworking supplies. - Cons: Potentially lower strength Option 2: - Pros: Higher strength - Cons: Heaver and is much harder to precisely cut.	**Option 1** was selected mainly because it was simple to precisely cut it into any required shape using the water jet cutter, allowing for efficient CAD design and quick manufacturing.
3D printing filament 	Option 1: PLA+ filament Option 2: Carbon Fibre Filament 	Option 1: - Pros: Cheaper and can be 3D printed on any 3d printer. - Cons: Relatively low strength Option 2: - Pros: Higher strength in the print direction - Cons: Much higher price	**Option 2** was selected for the final product due to its higher strength, which will be useful when the device is subjected to high loads. However, for early prototypes, Option 1 will be used.

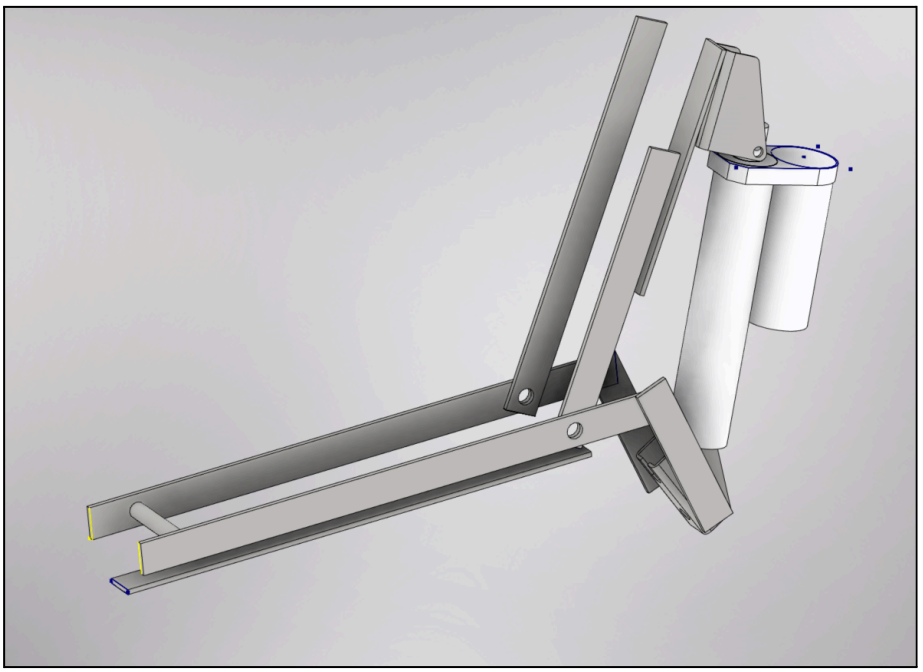
Prototype design evolution and selection

Option 1: Single sided mechanism



Option 1 was initially proposed due to its simplicity in manufacturing. It features a single lever made of a thin acrylic panel with a width of 3mm. At the end of the work arm of the lever, there is a 3D-printed circular cuff, allowing the force of the linear actuator to be transferred to the human wrist for strength enhancement and augmentation. The linear actuator is attached to two separate acrylic panels, and its extension causes a change in the angle of the arm. The mechanism rotates about a single pin, which is also made of acrylic or 3D-printed material. The device is intended to be attached to the upper arm using hook and loop velcro straps.However, after the initial 3D modelling of the prototype, several flaws were discovered. The primary flaw is the inability of the 3 mm acrylic material to withstand the forces exerted by the linear actuator and the load being lifted. Simulations showed that the mechanism would fail under the expected loads. Additionally, the attachment of the linear actuator to the acrylic sheets was another weak point, as the acrylic could not withstand such loads. Furthermore, the asymmetrical nature of the mechanism caused the forces exerted by the linear actuator and the load to be on different planes, inducing torsional strain. This torsional strain increased the risk of structural failure under load.

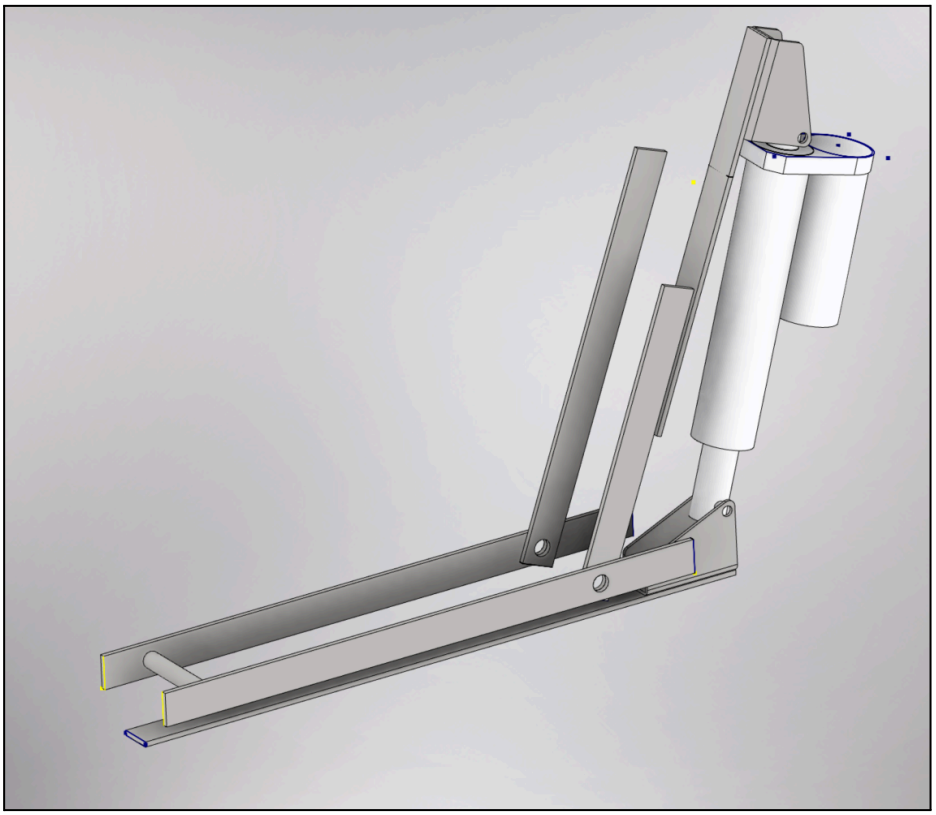
Option 2: Double sided mechanism (upper range of motion bias)



Option 2 was developed as a new approach to handling the forces exerted by the load upon the device. It features a lever mechanism on both sides of the human arm, rather than just on the anterior side, reducing the risk of torsional strain. The forces exerted by the linear actuator and the load are now on the same plane and are aligned. Additionally, a structural component has been fabricated out of steel rather than acrylic, significantly increasing the load-bearing capacity of the structural mechanism. Furthermore, steel brackets included with the linear actuator are being used to bear the force exerted by the linear actuator, further increasing reliability and serviceability.

The major drawback of this version of the device is that it biases the range of motion towards the higher end of what the human arm is capable of. This means that it is much more difficult to pick something up starting from an extended arm position, and the device cannot fully extend the human arm

Option 3: Double sided mechanism (lower range of motion bias)



Option 3 features many similar design choices compared to Option 2. However, the mechanism has been simplified, allowing the linear actuator to be attached in a different and more straightforward manner. This design allows the lower range of motion to be biassed rather than the upper range of motion, increasing practicality for lifting applications. The only major drawback of this adjustment is that it requires the linear actuator to be attached at a higher point on the arm, causing it to stick out above the shoulder of the user, which may impact the aesthetics of the device.

Design Choice and Justification

Option 3 was selected as the design to continue pursuing and improving upon due to its increased practicality and its ability to effectively bear the forces involved. The simplified attachment mechanism and the bias towards the lower range of motion make it more suitable for lifting applications. Despite the aesthetic drawback of the linear actuator sticking out above the shoulder, the overall functionality and structural integrity of Option 3 make it the preferred choice for further development.

Criteria 3: Planning the Creation of the System

Work Timeline Plan

Week	Tasks to be completed	Materials, components, processes/equipment
1	- Assembly of mechanical parts (steel & aluminium) and attachment of linear actuator - Completion of STL files for 3D printing, begin 3D printing	**Materials:** ● 300x300x8mm aluminium sheets, cut to correct sizes ● PLA+ filament **Components:** ● Linear actuator ● M8 bolts and nuts ● M10 bolts and nuts ● Control switch ● Kill switch ● Batteries ● Voltage regulator ● Motor controller ● Arduino **Processes/Equipment:** 1. Bracketing assorted aluminium parts together with M8 bolts and nuts a. Wrench (to tighten nuts) 2. Bracketing linear actuator with M8 bolts and nuts a. Wrench (to tighten nuts) 3. Cutting M10 bolt to correct length, adding aluminium parts, and securing nut using adhesive a. Hacksaw (to cut bolt) b. Cordless drill (to spin bolt) c. Hot-glue gun (for testing “welding” the nut) d. Soldering iron (for “welding” the nut) 4. Measure all of the electrical components and make sure that they can securely fit into the 3D model mechanism a. Digital callipers (for measuring) 5. Completing Inventor assembly of device and beginning to manufacture 3D printed components. a. 3D printer (additive manufacturing) b. Paint scraper (to remove 3D prints from print bed.
2	- Completion of wiring diagram - 3D components printed with prototype settings	**Materials:** ● PLA+ filament ● Wires **Components:** ● N/A **Processes/Equipment:** 1. Completing a full wiring diagram which

		includes all electrical components, with accurate colouring. 2. Completing Inventor assembly of device and beginning to manufacture 3D printed components. c. 3D printer (additive manufacturing) d. PLA+ filament (for prototyping) e. Paint scraper (to remove 3D prints from print bed.
3	- Adding wiring connectors (female to female) throughout the mechanism - Secure the switches, batteries, voltage regulator, and board (in an easily serviceable manner)	**Materials:** ● Heat shrink ● Hot glue ● Solder **Components:** ● Female to female jumper cables 100mm, assorted colours ● Control switch ● Kill switch ● Batteries ● Voltage regulator ● Motor controller ● Arduino **Processes/Equipment:** 1. Add female connectors to all of the components, like the switches, motor controller, to allow for easily replaceable components a. Soldering iron (for soldering and heat shrinking) b. Wire stripper (to strip wire insulation) c. Hot glue (used scarcely, to lightly hold components in place)
4	- Wire the circuit using male to male connectors as per the wiring diagram - Test for secure connections	**Materials:** ● Heat shrink ● Hot glue ● Solder **Components:** ● Male to male jumper cables 100mm, assorted colours **Processes/Equipment:** 1. Use male-male jumper cables to wire the control switch, kill switch, batteries, voltage regulator, motor controller, and arduino. a. Soldering iron (for permanent connections) b. Wire stripper 2. Check that all intended connections are secure and follow the wiring diagram a. Multimeter (using continuity test mode)
5	- Begin work on arduino program and “test as you go”	**Materials:** ● Arduino cable **Components:** ● N/A

		Processes/Equipment: 1. Create a pseudocode document of the logic behind the arduino code 2. Translate pseudocode document into C++ in arduino IDE 3. For each new function, test how it interacts with the device. 4. Continue until bare minimum features are functional
6	- Complete arduino program - Test device	**Materials:** ● Arduino cable **Components:** ● N/A **Processes/Equipment:** ● Continue with *Week 5, steps 2-4*
7	- Complete disassembly and reassembly, then test device	**Materials:** ● N/A **Components:** ● N/A **Processes/Equipment:** 1. Completely disassemble the device and reassemble device and record how long it takes, and any issues in testing afterwards a. Multimeter (for testing)
8	- Troubleshooting	*This is allocated as spare time to anticipate any unexpected challenges which delay any previous steps.*
9	- First human testing of device on light weights - Begin evaluation of device	**Materials:** ● Dumbbell weights (1-10kg) **Components:** ● N/A **Processes/Equipment:** 1. Begin evaluation of device as specified in later criteria
10	- Complete evaluation and documentation - Final submission	**Materials:** ● N/A **Components:** ● N/A **Processes/Equipment:** 1. Complete evaluation of device as specified in later criteria 2. Finish all documentation 3. Submit assignment

Soldering Iron

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment.
- Covered Footwear: Must be worn when operating this equipment.
- Hair Containment: Long and loose hair must be contained when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
- Ensure that risk assessment has been read.
2. Work Area Preparation:
- Check workspace and walkways to ensure no slip-hazards are present.
3. Equipment Tagging:
- Check electrical equipment is fitted with a current electrical safety test tag.
4. Examine Power Cord:
- Examine the power cord for damage.
5. Familiarisation:
- Locate and ensure you are familiar with the operation of the ON/OFF switch.
6. Tool Selection:
- Ensure the correct size tip is selected for the job.
7. Temperature Adjustment:
- Adjust the tip temperature to suit the required thermal transfer during soldering (nominally 340°C for tin/lead solder, 370°C for lead-free).
8. Material Safety Data Sheets (MSDS):
- Refer to relevant MSDS of all chemicals intended for use during the soldering process.
9. Faulty Equipment:
- Faulty equipment must not be used. Immediately report suspect equipment.

OPERATIONAL SAFETY CHECKS

1. Power Cord Handling:
- Never carry a power tool by the power cord or disconnect the tool by yanking the power cord.
2. Securing Workpiece:
- Ensure that the work piece is securely held or clamped whilst working.
3. Safe Positioning:
- Between operations, ensure the tool is rested safely to prevent inadvertent contact with skin or material.
4. Personal Safety:
- Use a safe working posture (beware of hair, skin or clothing in contact with hot tip).
5. Microscope Use:
- When using a stereo microscope, ensure the hot tip is in the lens field of view before looking into the microscope.
6. Avoid Contact:
- Avoid accidental contact with skin/hand.
7. Ventilation:
- Ensure adequate ventilation or, whenever possible, use fume extraction/filtration.
8. Solder Type:
- Whenever possible, use lead-free solder.
9. Cleaning:
- When using flux or solder paste, always clean the workpiece upon completion to remove residue.
10. Cooling:
- Clean tip before and after applying solder.

HOUSEKEEPING

1. Cooling Tools:
- Allow all tools and work pieces to cool down before handling or storing.
2. Proper Resting:
- Ensure the tool is rested on an appropriate stand or bench whilst cooling.
3. Switch Off:
- Switch off the tool when not in use.
4. Storage:
- Return all tooling and fixtures to the correct storage location.
5. Clean Area:
- Ensure the work area is left in a safe, clean and tidy state after the job is completed.
6. Surplus Material:
- Return surplus material to stock & dispose of waste material in an appropriate recycling or waste bin.
7. Hand Washing:
- Wash hands after handling solder wire, solder paste and/or flux.

POTENTIAL HAZARDS

- Burns from hot tool tips - Inhalation of fumes, particularly from tin/lead solder - Fire (Butane Gas Soldering Irons)	- Electric Shock - Eye injuries

FORBIDDEN

- Removal or modification of safety devices - Heating flammable/toxic materials, substances, plastics	- Changing Butane Gas cylinder refills near ignition source - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

3D Printer

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment.
- Covered Footwear: Must be worn when operating this equipment.
- Hair Containment: Long and loose hair must be contained when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Equipment Tagging:
   - Check electrical equipment is fitted with a current electrical safety test tag.
4. Examine Power Cord:
   - Examine the power cord for damage.
5. Familiarisation:
   - Locate and ensure you are familiar with the operation of the 3D printer.
6. Printer Settings:
   - Verify that the printer settings (temperature, speed, etc.) are appropriate for the material being used.
7. Material Safety Data Sheets (MSDS):
   - Refer to relevant MSDS of all materials intended for use during the printing process.
8. Faulty Equipment:
   - Faulty equipment must not be used. Immediately report suspect equipment.
9. Filament Check:
   - Ensure the correct type of filament is loaded and correctly fed into the printer.
10. Calibration:
    - Ensure the printer is properly calibrated for the print job.

OPERATIONAL SAFETY CHECKS

1. Power Cord Handling:
   - Never carry the printer by the power cord or disconnect the tool by yanking the power cord.
2. Printer Monitoring:
   - Ensure that the 3D printer is monitored during operation to prevent any issues.
3. Safe Positioning:
   - Maintain a safe distance from the printer while it is operating to avoid accidental contact.
4. Personal Safety:
   - Use a safe working posture and avoid wearing loose clothing or accessories.
5. Ventilation:
   - Ensure adequate ventilation or, whenever possible, use fume extraction/filtration.
6. Material Handling:
   - Handle all printing materials (filaments, resins) according to their safety instructions.
7. Cleaning:
   - Clean the print bed and nozzle before and after each print job to ensure optimal performance.
8. Cooling:
   - Allow printed objects to cool before handling.

HOUSEKEEPING

1. Cooling Tools:
   - Allow all tools and printed objects to cool down before handling or storing.
2. Proper Resting:
   - Ensure the printer is powered off and unplugged when not in use.
3. Switch Off:
   - Switch off the printer when not in use.
4. Storage:
   - Return all tools and materials to the correct storage location.
5. Clean Area:
   - Ensure the work area is left in a safe, clean and tidy state after the job is completed.
6. Material Disposal:
   - Dispose of waste material in an appropriate recycling or waste bin.
7. Hand Washing:
   - Wash hands after handling printing materials and finished prints.

POTENTIAL HAZARDS

- Burns from hot printer components (nozzle, bed) - Inhalation of fumes, particularly from certain filaments (ABS, resin) - Electric Shock	- Eye injuries - Fire risk from overheating

FORBIDDEN

- Removal or modification of safety devices - Heating flammable/toxic materials, substances, plastics	- Changing printer settings without proper knowledge - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Laser Cutter

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment, particularly if the laser cutter is not enclosed or if specified by the manufacturer.
- Covered Footwear: Must be worn when operating this equipment.
- Hair Containment: Long and loose hair must be contained when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Equipment Tagging:
   - Check electrical equipment is fitted with a current electrical safety test tag.
4. Examine Power Cord:
   - Examine the power cord for damage.
5. Familiarisation:
   - Locate and ensure you are familiar with the operation of the laser cutter, including emergency stop procedures.
6. Material Check:
   - Verify that the material being cut is suitable for laser cutting and free of hazardous coatings or adhesives.
7. Settings Verification:
   - Verify that the laser cutter settings (power, speed, focus) are appropriate for the material being used.
8. Ventilation:
   - Ensure that the ventilation system or fume extractor is operational.
9. Material Safety Data Sheets (MSDS):
   - Refer to relevant MSDS of all materials intended for use during the cutting process.
10. Faulty Equipment:
    - Faulty equipment must not be used. Immediately report suspect equipment.

OPERATIONAL SAFETY CHECKS

1. Power Cord Handling:
   - Never carry the tool by the power cord or disconnect the tool by yanking the power cord.
2. Monitoring:
   - Never leave the laser cutter unattended while it is operating.
3. Safe Positioning:
   - Maintain a safe distance from the laser cutter while it is operating to avoid accidental exposure to the laser beam.
4. Personal Safety:
   - Use a safe working posture and avoid wearing loose clothing or accessories.
5. Ventilation:
   - Ensure adequate ventilation or, whenever possible, use fume extraction/filtration.
6. Material Handling:
   - Handle all materials according to their safety instructions.
7. Cleaning:
   - Clean the laser cutter bed and optics before and after each job to ensure optimal performance.
8. Fire Safety:
   - Have a fire extinguisher nearby and know how to use it in case of an emergency.
9. Cooling:
   - Allow cut materials to cool before handling.

HOUSEKEEPING

1. Cooling Tools:
   - Allow all tools and cut materials to cool down before handling or storing.
2. Proper Resting:
   - Ensure the laser cutter is powered off and unplugged when not in use.
3. Switch Off:
   - Switch off the laser cutter when not in use.
4. Storage:
   - Return all tools and materials to the correct storage location.
5. Clean Area:
   - Ensure the work area is left in a safe, clean and tidy state after the job is completed.
6. Material Disposal:
   - Dispose of waste material in an appropriate recycling or waste bin.
7. Hand Washing:
   - Wash hands after handling materials and finishing cuts.

POTENTIAL HAZARDS

- Burns from hot cut materials - Inhalation of fumes, particularly from certain materials (plastics, resins) - Eye injuries from laser exposure	- Electric Shock - Fire risk from combustible materials

FORBIDDEN

- Removal or modification of safety devices - Cutting flammable/toxic materials without proper ventilation	- Changing laser cutter settings without proper knowledge - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

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Water-jet Cutter

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment.
- Covered Footwear: Must be worn when operating this equipment.
- Hearing Protection: Must be worn when operating this equipment due to high noise levels.
- Gloves: Must be worn when handling cut materials to prevent cuts and abrasions.
- Hair Containment: Long and loose hair must be contained when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Equipment Tagging:
   - Check electrical equipment is fitted with a current electrical safety test tag.
4. Examine Power and Water Lines:
   - Examine power cords and water lines for damage or leaks.
5. Familiarisation:
   - Locate and ensure you are familiar with the operation of the water jet cutter, including emergency stop procedures.
6. Material Check:
   - Verify that the material being cut is suitable for water jet cutting and free of hazardous coatings or adhesives.
7. Settings Verification:
   - Verify that the water jet cutter settings (pressure, speed, abrasive material) are appropriate for the material being used.
8. Water Supply:
   - Ensure the water supply is clean and adequate for the duration of the cutting operation.
9. Abrasive Material:
   - Ensure the appropriate abrasive material is loaded and the feed system is functioning correctly.
10. Faulty Equipment:
    - Faulty equipment must not be used. Immediately report suspect equipment.

OPERATIONAL SAFETY CHECKS

1. Power Cord Handling:
   - Never carry the tool by the power cord or disconnect the tool by yanking the power cord.
2. Monitoring:
   - Never leave the water jet cutter unattended while it is operating.
3. Safe Positioning:
   - Maintain a safe distance from the water jet cutter while it is operating to avoid accidental exposure to the high-pressure jet.
4. Personal Safety:
   - Use a safe working posture and avoid wearing loose clothing or accessories.
5. Ventilation:
   - Ensure adequate ventilation or, whenever possible, use fume extraction/filtration.
6. Material Handling:
   - Handle all materials according to their safety instructions.
7. Cleaning:
   - Clean the cutting bed and nozzles before and after each job to ensure optimal performance.
8. Fire Safety:
   - Have a fire extinguisher nearby and know how to use it in case of an emergency.
9. Cooling:
   - Allow cut materials to cool before handling.
10. Pressure Release:
    - Follow procedures for safely releasing any built-up pressure in the system before maintenance or adjustments.

HOUSEKEEPING

1. Cooling Tools:
   - Allow all tools and cut materials to cool down before handling or storing.
2. Proper Resting:
   - Ensure the water jet cutter is powered off and the water supply is turned off when not in use.
3. Switch Off:
   - Switch off the water jet cutter when not in use.
4. Storage:
   - Return all tools and materials to the correct storage location.
5. Clean Area:
   - Ensure the work area is left in a safe, clean and tidy state after the job is completed.
6. Material Disposal:
   - Dispose of waste material in an appropriate recycling or waste bin.
7. Hand Washing:
   - Wash hands after handling materials and finishing cuts.

POTENTIAL HAZARDS

- Cuts and abrasions from high-pressure water jet - Inhalation of dust or abrasive particles - Hearing damage from high noise levels	- Eye injuries from high-pressure jet or debris - Electric Shock - Slip hazards from water spills

FORBIDDEN

- Removal or modification of safety devices - Cutting hazardous materials without proper ventilation	- Changing water jet cutter settings without proper knowledge - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Wire Strippers

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from wire fragments.
- Gloves: Recommended to prevent cuts and abrasions when handling wires.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect wire strippers for damage or wear.
4. Familiarisation:
   - Ensure you are familiar with the operation of the wire strippers, including correct use techniques.
5. Material Check:
   - Verify that the wire to be stripped is suitable for the wire strippers being used.
6. Settings Verification:
   - Ensure the wire stripper is adjusted correctly for the wire gauge being stripped.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the wire strippers firmly and ensure a stable stance while operating.
2. Wire Positioning:
   - Position the wire properly in the stripping slot to avoid cutting the wire strands.
3. Stripping Motion:
   - Use a smooth stripping motion to avoid jerking, which can lead to injury.
4. Personal Safety:
   - Keep fingers clear of the stripping blades.
5. Cut Wire Disposal:
   - Properly dispose of wire insulation and fragments to maintain a clean workspace.
6. Tool Care:
   - Keep the wire strippers clean and free from grease or oil that could cause slipping.

HOUSEKEEPING

1. Clean Tools:
   - Clean the wire strippers after use to remove any wire fragments and residues.
2. Proper Storage:
   - Store wire strippers in a safe place to avoid damage and ensure they are readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of wire fragments after the job is completed.
4. Tool Inspection:
   - Regularly inspect wire strippers for wear and damage, and replace them if necessary.
5. Hand Washing:
   - Wash hands after handling wires and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges - Eye injuries from flying wire fragments	- Electric Shock (if working with live wires) - Strain injuries from improper use

FORBIDDEN

- Using wire strippers on live wires - Using damaged or worn-out wire strippers	- Modifying the tool or removing safety features - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Pliers

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from flying debris.
- Gloves: Recommended to prevent cuts and abrasions when handling materials.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect pliers for damage, rust, or wear.
4. Familiarisation:
   - Ensure you are familiar with the operation of the pliers, including correct use techniques.
5. Material Check:
   - Verify that the material to be handled is suitable for the pliers being used.
6. Settings Verification:
   - Ensure the pliers are adjusted correctly if they have adjustable settings.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the pliers firmly and ensure a stable stance while operating.
2. Material Positioning:
   - Position the material properly in the jaws of the pliers to avoid slippage.
3. Using Appropriate Force:
   - Apply the appropriate amount of force; do not use excessive force that may cause the pliers to slip or break.
4. Personal Safety:
   - Keep fingers clear of the jaws and pinch points.
5. Cut Material Disposal:
   - Properly dispose of any cut materials or debris to maintain a clean workspace.
6. Tool Care:
   - Keep the pliers clean and free from grease or oil that could cause slipping.

HOUSEKEEPING

1. Clean Tools:
   - Clean the pliers after use to remove any residues.
2. Proper Storage:
   - Store pliers in a safe place to avoid damage and ensure they are readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect pliers for wear and damage, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges - Eye injuries from flying debris	- Pinching injuries from improper use - Strain injuries from improper use

FORBIDDEN

- Using pliers on live wires unless they are insulated for electrical work - Using damaged or worn-out pliers	- Modifying the tool or removing safety features - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Hacksaw

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from flying debris.
- Gloves: Recommended to prevent cuts and abrasions when handling materials.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect hacksaw for damage, ensure the blade is securely fixed and in good condition.
4. Familiarisation:
   - Ensure you are familiar with the operation of the hacksaw, including correct sawing techniques.
5. Material Check:
   - Verify that the material to be cut is suitable for the hacksaw being used.
6. Settings Verification:
   - Ensure the blade is appropriate for the material being cut (e.g., metal or wood).
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the hacksaw firmly and ensure a stable stance while operating.
2. Material Positioning:
   - Secure the material properly in a vise or clamp to avoid movement during cutting.
3. Using Appropriate Force:
   - Apply consistent and even pressure; do not use excessive force that may cause the blade to break.
4. Cutting Technique:
   - Use smooth, controlled strokes; avoid jerky movements that could lead to accidents.
5. Personal Safety:
   - Keep fingers clear of the cutting path and blade.
6. Debris Disposal:
   - Properly dispose of any cut materials or debris to maintain a clean workspace.
7. Tool Care:
   - Keep the hacksaw clean and free from grease or oil that could cause slipping.

HOUSEKEEPING

1. Clean Tools:
   - Clean the hacksaw after use to remove any residues.
2. Proper Storage:
   - Store hacksaw in a safe place to avoid damage and ensure it is readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the hacksaw for wear and damage, and replace the blade if necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges - Eye injuries from flying debris	- Blade breakage leading to injury - Strain injuries from improper use

FORBIDDEN

- Using a damaged or worn-out hacksaw or blade - Modifying the tool or removing safety features	- Using excessive force that could lead to blade breakage - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Linear Actuator

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from flying debris.
- Gloves: Recommended to prevent cuts and abrasions when handling components.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect the linear actuator for damage, ensure all components are securely fixed and in good condition.
4. Electrical Check:
   - Verify that the electrical connections and power supply are in good condition and properly rated for the actuator.
5. Familiarisation:
   - Ensure you are familiar with the operation of the linear actuator, including control mechanisms and emergency stop procedures.
6. Material Check:
   - Verify that the materials to be moved or manipulated are suitable for the actuator's capacity and range.
7. Settings Verification:
   - Ensure the actuator settings (speed, travel distance, etc.) are appropriate for the task.
8. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Handle the actuator and its controls carefully, ensuring stable and secure operation.
2. Material Positioning:
   - Secure the material properly to the actuator to avoid movement or slippage during operation.
3. Using Appropriate Force:
   - Do not exceed the actuator's rated capacity; apply loads within its specified limits.
4. Movement Awareness:
   - Be aware of the actuator's range of motion and ensure no body parts or obstructions are within this range.
5. Personal Safety:
   - Keep fingers and other body parts clear of the actuator and moving parts.
6. Emergency Stop:
   - Be familiar with and ready to use the emergency stop mechanism in case of malfunction or unexpected movement.
7. Tool Care:
   - Keep the actuator clean and free from debris that could interfere with its operation.

HOUSEKEEPING

1. Clean Tools:
   - Clean the actuator and surrounding area after use to remove any residues.
2. Proper Storage:
   - Store the actuator in a safe place to avoid damage and ensure it is readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the actuator for wear and damage, and perform maintenance as necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges - Eye injuries from flying debris	- Strain injuries from improper use - Mechanical failure leading to sudden or unexpected movement

FORBIDDEN

- Using a damaged or malfunctioning linear actuator - Modifying the actuator or removing safety features	- Exceeding the actuator's rated capacity - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Lithium-ion Battery

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when handling batteries to protect from splashes or debris.
- Gloves: Must be worn to protect hands from chemicals and electrical hazards.
- Fire-Resistant Lab Coat: Recommended to minimise risk from battery fires.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
   - Ensure the area is free from flammable materials.
3. Tool Inspection:
   - Inspect batteries for damage, swelling, or leakage.
   - Ensure batteries are stored in a cool, dry place away from direct sunlight and heat sources.
4. Charging Equipment Check:
   - Verify that the charging equipment is in good condition and suitable for the battery type.
5. Familiarisation:
   - Ensure you are familiar with the operation and handling of lithium-ion batteries, including proper charging and discharging procedures.
6. Material Check:
   - Verify that all materials and tools are suitable for working with lithium-ion batteries.
7. Emergency Preparedness:
   - Ensure fire extinguishers (Class D for lithium fires) are nearby and know the procedures for handling battery fires.
8. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Handle batteries carefully to avoid puncturing, crushing, or dropping them.
2. Battery Usage:
   - Use batteries within their specified limits and avoid overcharging or deep discharging.
3. Temperature Monitoring:
   - Monitor battery temperature during use and charging to prevent overheating.
4. Charging Practices:
   - Only use chargers designed for the specific type of lithium-ion battery.
5. Personal Safety:
   - Keep batteries away from conductive materials to prevent short circuits.
6. Ventilation:
   - Ensure the work area is well-ventilated to avoid accumulation of any hazardous fumes.
7. Disposal of Damaged Batteries:
   - Properly dispose of damaged or defective batteries in accordance with local regulations.
8. Tool Care:
   - Keep batteries and charging equipment clean and free from dust and debris.

HOUSEKEEPING

1. Clean Tools:
   - Clean batteries and surrounding area after use to remove any residues.
2. Proper Storage:
   - Store batteries in a safe place, preferably in a fire-resistant container, to avoid damage and ensure they are readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect batteries and charging equipment for wear and damage, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling batteries and tools to remove any residues.

POTENTIAL HAZARDS

- Fire and explosion from short circuits or overheating - Chemical burns from leaking electrolytes - Electric shock from improper handling	- Inhalation of toxic fumes from damaged batteries - Environmental contamination from improper disposal

FORBIDDEN

- Using damaged or swollen batteries - Using inappropriate chargers or charging methods - Exposing batteries to extreme temperatures or direct sunlight - Disassembling or puncturing batteries	- Modifying the battery or removing safety features - Disposing of batteries in regular trash - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Cordless Drill

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from flying debris.
- Gloves: Recommended to prevent cuts and abrasions when handling materials.
- Hearing Protection: Recommended to protect against noise exposure.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect the cordless drill for damage or wear, including the battery, chuck, and drill bits.
4. Battery Check:
   - Ensure the battery is fully charged and properly installed.
5. Familiarisation:
   - Ensure you are familiar with the operation of the cordless drill, including speed settings and reverse function.
6. Material Check:
   - Verify that the material to be drilled is suitable for the drill bit being used.
7. Settings Verification:
   - Ensure the drill settings (speed, torque) are appropriate for the task.
8. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the drill firmly with both hands and ensure a stable stance while operating.
2. Material Positioning:
   - Secure the material properly in a vise or clamp to avoid movement during drilling.
3. Drill Bit Installation:
   - Ensure the drill bit is correctly and securely installed in the chuck.
4. Using Appropriate Force:
   - Apply consistent and even pressure; do not force the drill bit through the material.
5. Personal Safety:
   - Keep fingers and other body parts clear of the drilling area.
6. Debris Management:
   - Properly manage and dispose of any debris or shavings to maintain a clean workspace.
7. Battery Management:
   - Monitor the battery level and switch to a fully charged battery when needed.
8. Emergency Stop:
   - Be familiar with and ready to use the emergency stop mechanism in case of malfunction or unexpected movement

HOUSEKEEPING

1. Clean Tools:
   - Clean the cordless drill and drill bits after use to remove any residues.
2. Proper Storage:
   - Store the drill and accessories in a safe place to avoid damage and ensure they are readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the drill, battery, and bits for wear and damage, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges and drill bits - Eye injuries from flying debris - Electric shock from battery malfunction	- Noise-induced hearing loss from prolonged use - Strain injuries from improper use

FORBIDDEN

- Using a damaged or malfunctioning cordless drill - Using inappropriate or damaged drill bits - Exceeding the drill's rated capacity	- Modifying the tool or removing safety features - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Paint scraper

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn when operating this equipment to protect from flying debris.
- Gloves: Must be worn to prevent cuts and abrasions.
- Covered Footwear: Must be worn when operating this equipment.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect the scraper tool for damage or wear, including the blade.
4. Blade Check:
   - Ensure the blade is sharp and securely attached.
5. Familiarisation:
   - Ensure you are familiar with the operation of the scraper tool, including proper scraping techniques.
6. Material Check:
   - Verify that the material to be scraped is suitable for the scraper tool being used.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the scraper tool firmly and ensure a stable stance while operating.
2. Scraping Technique:
   - Use a consistent and controlled motion; do not use excessive force that may cause the blade to slip or break.
3. Personal Safety:
   - Keep fingers and other body parts clear of the blade.
4. Debris Management:
   - Properly manage and dispose of any debris to maintain a clean workspace.
5. Tool Care:
   - Keep the scraper tool clean and free from grease or oil that could cause slipping.

HOUSEKEEPING

1. Clean Tools:
   - Clean the scraper tool after use to remove any residues.
2. Proper Storage:
   - Store the scraper tool in a safe place to avoid damage and ensure it is readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the scraper tool for wear and damage, and replace the blade if necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges and blades - Eye injuries from flying debris	- Strain injuries from improper use

FORBIDDEN

- Using a damaged or dull scraper tool - Using excessive force that could lead to blade breakage - Modifying the tool or removing safety features	- Distracting operator - Using equipment with lapsed test tag

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Digital Calipers

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Recommended to protect from any accidental splashes or debris.
- Gloves: Recommended to prevent cuts and abrasions when handling sharp or rough materials.
- Covered Footwear: Must be worn to protect from falling objects.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect the digital callipers for any visible damage or wear.
   - Check that the battery is functioning and that the display is clear.
4. Calibration Check:
   - Verify that the callipers are properly calibrated before use.
5. Familiarisation:
   - Ensure you are familiar with the operation of the digital callipers, including how to zero the tool and switch between measurement units.
6. Material Check:
   - Verify that the material to be measured is suitable for the callipers.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during measurement.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Handle the callipers carefully to avoid dropping or damaging the tool.
2. Measurement Technique:
   - Ensure the callipers are properly zeroed before taking measurements.
   - Use a consistent and controlled motion when measuring; do not force the tool onto the material.
3. Personal Safety:
   - Keep fingers clear of the measuring jaws to avoid pinching injuries.
4. Reading the Display:
   - Ensure the display is clean and readable to avoid misinterpretation of measurements.
5. Tool Care:
   - Avoid exposing the callipers to moisture, dust, or excessive force that could affect accuracy.

HOUSEKEEPING

1. Clean Tools:
   - Clean the digital callipers after use to remove any residues.
2. Proper Storage:
   - Store the callipers in their protective case to avoid damage and ensure they are readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the callipers for wear and damage, and replace the battery as necessary.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Pinching injuries from the measuring jaws - Inaccurate measurements due to tool damage or improper use	- Battery leakage causing chemical exposure

FORBIDDEN

- Using damaged or malfunctioning digital callipers - Applying excessive force that could damage the tool or the material	- Modifying the tool or removing safety features - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Wrench

DO NOT use this component unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this component.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Recommended to protect from any accidental splashes or debris.
- Gloves: Recommended to prevent cuts and abrasions when handling materials.
- Covered Footwear: Must be worn to protect from falling objects.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards are present.
3. Tool Inspection:
   - Inspect the wrench for any visible damage or wear, ensuring it is the correct size and type for the task.
4. Tool Selection:
   - Verify that the wrench is appropriate for the fastener type and size.
5. Familiarisation:
   - Ensure you are familiar with the proper use of the wrench, including correct hand positioning and application of force.
6. Material Check:
   - Verify that the material being worked on is stable and properly supported.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the wrench firmly and ensure a stable stance while operating.
2. Fastener Positioning:
   - Ensure the wrench is properly seated on the fastener to avoid slippage.
3. Using Appropriate Force:
   - Apply consistent and even pressure; do not use excessive force that may cause the wrench to slip or the fastener to strip.
4. Personal Safety:
   - Keep fingers and other body parts clear of the wrench's path.
5. Tool Care:
   - Keep the wrench clean and free from grease or oil that could cause slipping.
6. Torque Awareness:
   - Be aware of the appropriate torque specifications for the fastener to prevent over-tightening or under-tightening.

HOUSEKEEPING

1. Clean Tools:
   - Clean the wrench after use to remove any residues.
2. Proper Storage:
   - Store the wrench in a safe place to avoid damage and ensure it is readily available for future use.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris after the job is completed.
4. Tool Inspection:
   - Regularly inspect the wrench for wear and damage, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling

POTENTIAL HAZARDS

- Cuts and abrasions from sharp edges or slipping wrenches - Strain injuries from improper use or excessive force	- Damage to fasteners or materials from incorrect wrench use

FORBIDDEN

- Using a damaged or malfunctioning wrench - Using inappropriate or incorrect size/type of wrench for the fastener - Applying excessive force that could lead to tool slippage or fastener damage	- Modifying the tool or removing safety features - Distracting operator

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Heat Gun

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn to protect from potential sparks, debris, or splashes.
- Heat-Resistant Gloves: Must be worn to protect from burns.
- Long-Sleeved Clothing: Recommended to protect skin from exposure to high heat.
- Covered Footwear: Must be worn to protect from falling objects or hot materials.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Check workspace and walkways to ensure no slip-hazards or flammable materials are present.
3. Tool Inspection:
   - Inspect the heat gun for any visible damage, wear, or frayed cables.
   - Ensure the nozzle is clean and unobstructed.
4. Power Source Check:
   - Ensure the power source is appropriate and safely accessible.
5. Familiarisation:
   - Ensure you are familiar with the operation of the heat gun, including temperature controls and proper use techniques.
6. Material Check:
   - Verify that the material to be heated is suitable for the temperature setting and won't produce harmful fumes.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the heat gun securely and maintain a firm grip at all times.
   - Ensure the heat gun is always pointed away from yourself and others.
2. Adjusting Temperature:
   - Set the temperature and airflow to the correct level for the material being worked on.
   - Avoid overheating materials that can burn or release toxic fumes.
3. Distance Management:
   - Maintain a safe working distance between the heat gun and the material being heated.
4. Ventilation:
   - Ensure the workspace is properly ventilated to avoid inhalation of fumes, especially when working with plastics, paint, or adhesives.
5. Continuous Monitoring:
   - Continuously monitor the material being heated to prevent burns, warping, or accidental fires.
6. Fire Safety:
   - Keep a fire extinguisher or other fire suppression device nearby in case of ignition.

HOUSEKEEPING

1. Clean Tools:
   - Allow the heat gun to cool completely after use before cleaning or storing.
2. Proper Storage:
   - Store the heat gun in a safe place, away from flammable materials and out of reach of children.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of debris or any materials that may have been affected by heat.
4. Tool Inspection:
   - Regularly inspect the heat gun for wear and damage, especially the nozzle and cord.
5. Hand Washing:
   - Wash hands after handling materials and tools to remove any residues.

POTENTIAL HAZARDS

- Burns from direct exposure to the heat gun nozzle or heated materials - Fire hazards from overheating or ignition of nearby flammable materials - Eye injuries from flying debris, sparks, or fumes	- Inhalation of harmful fumes from heated materials - Electrical shock if the tool is damaged or improperly handled

FORBIDDEN

- Using a damaged or malfunctioning heat gun - Leaving the heat gun unattended while powered on or hot - Pointing the heat gun at yourself or others - Overheating materials beyond their recommended temperature	- Using in a poorly ventilated area - Modifying the tool or removing safety features - Distracting operator - Using equipment with lapsed test tag

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

12V Power Supply

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn to protect from potential sparks or short circuits.
- Rubber-Soled Footwear: Recommended to protect against electrical shock.
- Insulated Gloves: Recommended when handling live wires or electrical components.
- Fire-Resistant Clothing: Recommended to reduce the risk of burns from sparks or accidental electrical fires.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Ensure that the workspace is dry, free of water, and well-organised with no slip-hazards.
3. Tool Inspection:
   - Inspect the 12V power supply for visible damage, frayed cables, or exposed wiring.
   - Ensure that the output voltage is stable and correct.
4. Connection Inspection:
   - Verify that all connections, terminals, and wiring are secure and properly insulated.
5. Power Source Check:
   - Ensure the power supply is connected to a properly grounded power source.
6. Familiarisation:
   - Ensure you are familiar with the operation of the power supply, including input/output voltage settings and overload protection features.
7. Material Check:
   - Ensure the load connected to the power supply is rated for 12V operation and has no damage.
8. Proper Lighting:
   - Ensure the work area is well-lit to avoid mistakes during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Handle the power supply and connected equipment with care, ensuring wires are kept untangled and organised.
2. Voltage Check:
   - Double-check that the voltage and polarity settings are correct for the device or circuit connected to the power supply.
3. Avoid Overloading:
   - Do not connect loads that exceed the power supply's rated current or wattage.
4. Ventilation:
   - Ensure the power supply is placed in a well-ventilated area to prevent overheating.
5. Electrical Isolation:
   - Isolate the power supply before making any changes to wiring or connections to avoid electric shock or damage.
6. Avoid Contact:
   - Never touch exposed wires or terminals while the power supply is on, and avoid direct contact with live components.
7. Fire Safety:
   - Keep a fire extinguisher nearby in case of electrical fires.
8. Emergency Shutdown:
   - Be familiar with and ready to use the emergency shut-off in case of malfunction or electrical hazards.

HOUSEKEEPING

1. Clean Tools:
   - Disconnect the power supply from the mains and allow it to cool before cleaning.
   - Wipe the power supply clean with a dry cloth to remove any dust or dirt.
2. Proper Storage:
   - Store the power supply in a cool, dry place, away from flammable materials and out of reach of children.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of loose wiring or debris that may present electrical hazards.
4. Tool Inspection:
   - Regularly inspect the power supply for wear, loose connections, or damaged components, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling electrical components and wires to remove any residues.

POTENTIAL HAZARDS

- Electric shock from live wires or terminals
- Short circuits causing sparks or fire
- Burns from overheating components or wiring
- Damage to connected devices from incorrect voltage or polarity
- Fire hazards due to overloading or malfunctioning power supply

FORBIDDEN

- Using a damaged or malfunctioning power supply
- Exceeding the power supply's rated capacity
- Connecting live wires without powering off the unit
- Touching exposed wires or terminals while powered on
- Modifying the tool or removing safety features
- Distracting operator
- Using equipment with lapsed test tag

This SOP does not necessarily cover all possible hazards and should be used in conjunction with other references. It is designed to be used as an adjunct to teaching safety procedures and act as a reminder prior to machine use.

Hot Glue Gun

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Recommended to protect from splashes of hot glue.
- Heat-Resistant Gloves: Recommended to prevent burns from hot glue and the heated nozzle.
- Long-Sleeved Clothing: Recommended to protect skin from exposure to hot glue.
- Covered Footwear: Must be worn to protect from falling objects or hot glue drops.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Ensure that the workspace is clean, dry, and free of flammable materials or slip hazards.
3. Tool Inspection:
   - Inspect the hot glue gun for visible damage, especially to the cord and nozzle.
   - Ensure the glue sticks are appropriate for the tool's temperature rating.
4. Power Source Check:
   - Ensure the power source is appropriate and safely accessible.
5. Familiarisation:
   - Ensure you are familiar with the operation of the hot glue gun, including how to load glue sticks, heat the tool, and control the flow of glue.
6. Material Check:
   - Verify that the material to be glued is suitable for hot glue adhesion and won't be damaged by heat.
7. Proper Lighting:
   - Ensure the work area is well-lit to avoid errors during operation.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the glue gun securely and ensure a stable stance while operating.
   - Keep fingers clear of the nozzle to prevent burns.
2. Heating Time:
   - Allow the glue gun to fully heat up before use, typically for a few minutes, as indicated by the manufacturer.
3. Application Technique:
   - Apply the glue in a controlled manner, ensuring the glue gun is properly angled for precise application.
4. Personal Safety:
   - Avoid touching the hot glue or the nozzle to prevent burns.
   - Use a work surface that can withstand heat and prevent hot glue from dripping onto your skin or clothing.
5. Ventilation:
   - Ensure the workspace is well-ventilated to avoid inhalation of fumes, especially when using the hot glue gun on plastics or other materials that may emit vapors.
6. Fire Safety:
   - Keep a fire extinguisher or other fire suppression device nearby in case of ignition.
7. Tool Placement:
   - Place the glue gun on its stand or a heat-resistant surface when not in use; never leave it lying on the workbench or near flammable materials.

HOUSEKEEPING

1. Clean Tools:
   - Unplug the glue gun and allow it to cool completely before cleaning.
   - Wipe off any residual glue from the nozzle using a cloth (when cool).
2. Proper Storage:
   - Store the glue gun in a safe place, away from flammable materials and out of reach of children.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of glue residues or debris that may present hazards.
4. Tool Inspection:
   - Regularly inspect the glue gun for wear, loose parts, or damaged cords, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling glue sticks or any material to remove residues.

POTENTIAL HAZARDS

- Burns from contact with hot glue or the nozzle - Fire hazards from leaving the glue gun unattended or near flammable materials - Eye injuries from hot glue splashes	- Inhalation of fumes from melted plastic or other hazardous materials - Electric shock from damaged cords or improper use

FORBIDDEN

- Using a damaged or malfunctioning glue gun - Leaving the glue gun unattended while powered on or hot - Pointing the glue gun at yourself or others - Using the glue gun on inappropriate materials (e.g., highly flammable materials)	- Modifying the tool or removing safety features - Distracting operator - Using equipment with lapsed test tag

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Multimeter

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn to protect from potential sparks or short circuits.
- Insulated Gloves: Recommended when working with high voltage or live circuits.
- Rubber-Soled Footwear: Must be worn to prevent electric shock.
- Non-conductive Clothing: Avoid wearing metal jewellery or conductive materials while operating the multimeter.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
   - Specific risk assessment task ID (e.g., UQ risk assessment task ID #XXXXX).
2. Work Area Preparation:
   - Ensure the workspace is dry, clean, and free of water or conductive materials.
   - Ensure no flammable materials are nearby when testing live circuits.
3. Tool Inspection:
   - Inspect the multimeter for visible damage, such as frayed test leads or cracked casing.
   - Ensure the batteries are properly installed and charged.
4. Familiarisation:
   - Ensure you are familiar with the correct operation of the multimeter, including the settings for measuring voltage, current, resistance, and continuity.
   - Understand the maximum voltage and current ratings of the multimeter.
5. Material Check:
   - Verify that the circuit or equipment being tested is compatible with the multimeter's capabilities and does not exceed its rating.
6. Proper Lighting:
   - Ensure the work area is well-lit to easily read the display and correctly connect the probes.

OPERATIONAL SAFETY CHECKS

1. Tool Handling:
   - Hold the glue gun securely and ensure a stable stance while operating.
   - Keep fingers clear of the nozzle to prevent burns.
2. Heating Time:
   - Allow the glue gun to fully heat up before use, typically for a few minutes, as indicated by the manufacturer.
3. Application Technique:
   - Apply the glue in a controlled manner, ensuring the glue gun is properly angled for precise application.
4. Personal Safety:
   - Avoid touching the hot glue or the nozzle to prevent burns.
   - Use a work surface that can withstand heat and prevent hot glue from dripping onto your skin or clothing.
5. Ventilation:
   - Ensure the workspace is well-ventilated to avoid inhalation of fumes, especially when using the hot glue gun on plastics or other materials that may emit vapors.
6. Fire Safety:
   - Keep a fire extinguisher or other fire suppression device nearby in case of ignition.
7. Tool Placement:
   - Place the glue gun on its stand or a heat-resistant surface when not in use; never leave it lying on the workbench or near flammable materials.

HOUSEKEEPING

1. Clean Tools:
   - Unplug the glue gun and allow it to cool completely before cleaning.
   - Wipe off any residual glue from the nozzle using a cloth (when cool).
2. Proper Storage:
   - Store the glue gun in a safe place, away from flammable materials and out of reach of children.
3. Work Area Cleaning:
   - Ensure the work area is left clean and free of glue residues or debris that may present hazards.
4. Tool Inspection:
   - Regularly inspect the glue gun for wear, loose parts, or damaged cords, and replace if necessary.
5. Hand Washing:
   - Wash hands after handling glue sticks or any material to remove residues.

POTENTIAL HAZARDS

- Burns from contact with hot glue or the nozzle - Fire hazards from leaving the glue gun unattended or near flammable materials - Eye injuries from hot glue splashes	- Inhalation of fumes from melted plastic or other hazardous materials - Electric shock from damaged cords or improper use

FORBIDDEN

- Using a damaged or malfunctioning glue gun - Leaving the glue gun unattended while powered on or hot - Pointing the glue gun at yourself or others - Using the glue gun on inappropriate materials (e.g., highly flammable materials)	- Modifying the tool or removing safety features - Distracting operator - Using equipment with lapsed test tag

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Mallet

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn to protect eyes from flying debris or fragments.
- Gloves: Recommended to protect hands from splinters, vibrations, and accidental impact.
- Steel-Toe Boots: Recommended to protect feet from dropped tools or materials.
- Long-Sleeved Clothing: To protect arms from accidental contact with sharp or hard surfaces.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Ensure the workspace is clean, dry, and free of obstructions or trip hazards.
   - Remove any materials that could cause slipping or distractions.
3. Tool Inspection:
   - Inspect the mallet for damage, such as cracks or chips in the head or handle.
   - Ensure that the head is securely attached to the handle and there is no looseness.
4. Familiarisation:
   - Ensure you are familiar with the proper handling and striking techniques.
5. Material Check:
   - Verify that the material being struck is appropriate for a mallet and that it is securely clamped or held in place.
6. Proper Lighting:
   - Ensure the work area is well-lit to allow for accurate, controlled strikes.

OPERATIONAL SAFETY CHECKS

1. Proper Grip:
   - Hold the mallet securely with both hands, ensuring a firm grip to avoid slipping during use.
2. Positioning:
   - Stand in a stable position with feet shoulder-width apart to maintain balance during strikes.
3. Controlled Strikes:
   - Use controlled, accurate strikes to avoid injury or damage to the workpiece.
   - Always strike away from your body and maintain a safe distance from others.
4. Avoid Overstriking:
   - Do not use excessive force, as overstriking may cause loss of control, damage to the tool, or material.
5. Hand and Body Position:
   - Keep hands, fingers, and other body parts clear of the area where you are striking.
6. Maintenance:
   - Periodically check the mallet for signs of wear or damage during use.
7. Fire Safety:
   - Ensure the mallet is not used near open flames or hot surfaces, particularly if the handle is made of wood.

HOUSEKEEPING

1. Clean Tools:
After use, wipe the mallet clean of any dirt, dust, or debris.
2. Proper Storage:
Store the mallet in a safe, designated area, away from moisture, to prevent wood handles from warping or metal heads from rusting.
3. Work Area Cleaning:
Ensure the work area is left clean and free of debris or materials that could pose a hazard.
4. Tool Inspection:
Regularly inspect the mallet for wear or damage, especially cracks in the handle or head. Replace if necessary.
5. Hand Washing:
Wash hands after handling materials or tools, especially if working with adhesives, sealants, or other substances.

POTENTIAL HAZARDS

- Striking injuries to hands, arms, or feet from uncontrolled or poorly aimed blows - Flying debris or fragments causing eye injuries - Muscle strain or injury from improper handling techniques	- Damage to materials from excessive force - Dropped tools potentially injuring feet or others

FORBIDDEN

- Using a damaged or malfunctioning mallet - Using excessive force or overstriking - Using the mallet for purposes other than its intended use (e.g., striking metal chisels)	- Striking with the mallet in an unstable or crowded area - Distracting operator - Modifying the tool or removing safety features

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Sandpaper

DO NOT use this tool unless you have been trained and assessed to a competent level in its safe use and operation, and have been given permission to use this equipment.

PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIREMENTS

- Safety Glasses: Must be worn to protect eyes from dust or debris.
- Dust Mask/Respirator: Recommended to avoid inhaling fine dust particles.
- Gloves: Recommended to protect hands from abrasions and splinters.
- Hearing Protection: Recommended when using sandpaper in combination with powered tools like sanders.
- Long-Sleeved Clothing: To protect arms from contact with rough surfaces or debris.

PRE-OPERATIONAL SAFETY CHECKS

1. Risk Assessment:
   - Ensure that risk assessment has been read.
2. Work Area Preparation:
   - Ensure the workspace is clean, dry, and free from obstructions or slip hazards.
   - Set up proper dust collection or ventilation systems to reduce airborne dust.
3. Material Check:
   - Ensure the material being sanded is secured in place to prevent movement during sanding.
   - Verify that the material is appropriate for sanding (e.g., wood, metal, etc.).
4. Tool Inspection:
   - Inspect the sandpaper for damage, such as tears or worn-out areas.
   - Choose the appropriate grit of sandpaper for the task at hand.
5. Proper Lighting:
   - Ensure the work area is well-lit to monitor the progress and surface quality.

OPERATIONAL SAFETY CHECKS

1. Proper Grip:
   - Hold the sandpaper securely with firm, even pressure, or use a sanding block for better control.
2. Controlled Movements:
   - Use smooth, even strokes in the direction of the grain (when sanding wood) or appropriate technique for the material being worked on.
3. Dust Control:
   - Ensure that proper ventilation is in place and dust extraction systems are active to reduce the inhalation of fine particles.
   - Avoid blowing dust off the workpiece, which can spread particles into the air.
4. Frequent Breaks:
   - Take regular breaks when sanding to inspect progress and prevent muscle strain or repetitive motion injuries.
5. Avoid Over-Sanding:
   - Do not apply excessive force, as it may damage the workpiece or lead to unnecessary wear on the sandpaper.
6. Use of Powered Tools:
   - If using sandpaper with powered tools (e.g., orbital sander), ensure the tool is in good working order, and follow its specific safety guidelines.

HOUSEKEEPING

1. Clean Tools:
   - After use, clean the sandpaper by gently tapping it to remove excess dust or debris.
2. Proper Storage:
   - Store sandpaper in a dry, cool location to prevent it from deteriorating due to moisture.
3. Work Area Cleaning:
   - Clean the work area to remove sanding dust, which can pose a fire or respiratory hazard.
   - Use a vacuum or dust collection system to capture fine dust particles.
4. Tool Inspection:
   - Regularly inspect sandpaper for wear and replace it when it becomes too smooth or torn.
5. Hand Washing:
   - Wash hands after handling sandpaper and materials to remove dust and residue.

POTENTIAL HAZARDS

- Eye irritation or injury from airborne dust particles - Respiratory issues due to inhaling fine dust - Skin abrasions or cuts from rough surfaces or sharp material edges	- Muscle strain or repetitive motion injuries from prolonged sanding - Damage to materials from using the wrong grit or over-sanding - Fire hazards when sanding flammable materials in dusty environments

FORBIDDEN

- Using damaged or excessively worn sandpaper - Sanding without securing the material properly - Applying excessive force while sanding - Ignoring dust control systems or ventilation when working in enclosed areas	- Using sandpaper near open flames or heat sources (risk of fire from dust) - Distracting operator - Sanding without appropriate PPE, especially a dust mask or respirator

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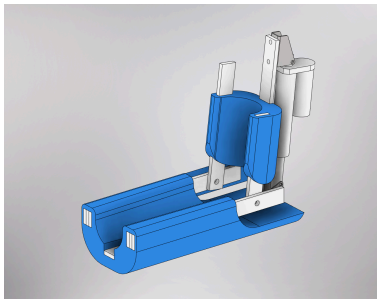
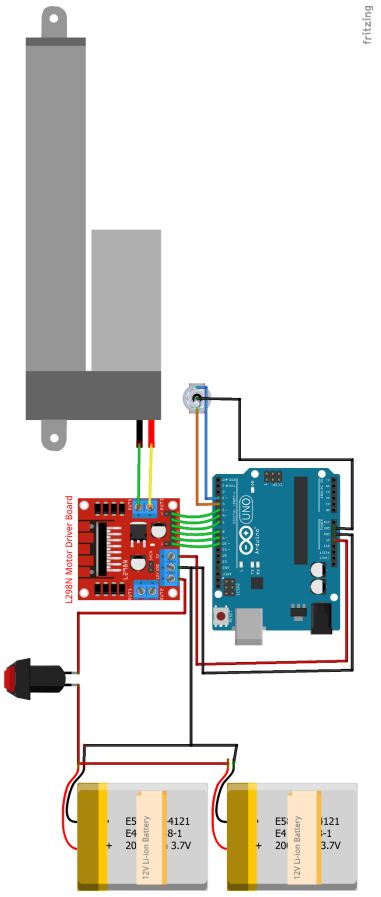
Criteria 4, 5 & 7 : Development Log

This development log outlines the processes, tools, equipment, components, materials, and OH&S considerations involved in the production of the exoARM device.

Week 1

Day 1:

Image	Outline	Explanation
	**Processes** • CAD Design **Tools/Equipment/Software** • Autodesk Inventor	After finalising the design of the metal components of the device (shown in white), the next step was to create a design for the plastic 3D printed components which exist to connect the metal beams and to aid in comfort and load distribution. The image shows the first iteration of the 3D printed parts for the device, with one large part for the lower arm and a smaller part for the upper arm.
	**Processes** • Circuit Design **Tools/Equipment/Software** • Fritzing	A circuit diagram was completed for the device. The diagram includes two 12V batteries (far left), Arduino Uno R3 and L298N (centre), and a 1500N linear actuator (right). Control systems are also present in the diagram, including an emergency OFF switch for the batteries (SPST normally closed push button), as well as a rocker switch for controlling the extension and retraction of the device (ON-OFF-ON rocker switch). The batteries are wired in parallel and connected to the L298N 12V input. The ground from the L298N is connected to an Arduino GND and the battery ground wire. Any other connections are clearly shown in the diagram.

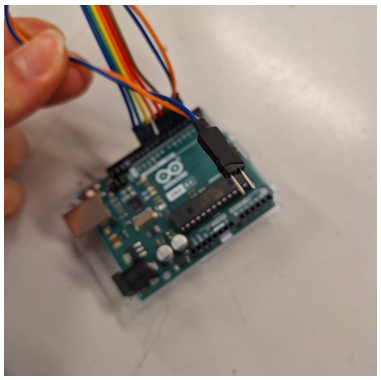
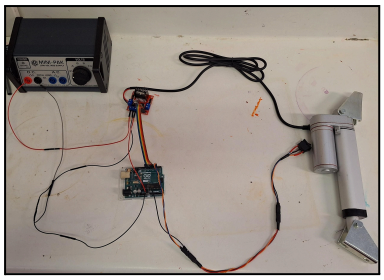
Notes and next steps:

For the 3D printed parts, the following additions need to be made to the 3D model ASAP before printing the parts:

- Channels for routing wiring through the assembly, to prevent the wiring from exposure to external conditions and damage. (this change is addressed on Week 5, Day 1)
- A enclosure with a control mechanism for the device, as all of the components will not fit on the device. This should be robust and able to be put in a backpack for use friendliness. (this change is addressed on Week 7, Day 3)
- Additionally, bolt holes need to be added for securing the steel components to the device. Tolerances must be considered for this. (this change is addressed on Week 5, Day 1)

Day 2:

Today I started wiring the components together, in order to test them independently of the mechanism. I used Dupont connectors so that the components can be hot swappable in case of failure, improving serviceability.

Image	Outline	Explanation
	**Processes** • Heat Shrinking **Tools/Equipment/Software** • Heat Gun **Materials involved** • Jumper wires with dupont connectors • 3mm diameter heat shrink **OH&S considerations** • Risk Assessment for “Heat Gun”	This image shows two dupont connectors connected with heat shrink. For ease of wiring, any associated connectors (such as the two input pins for the rocker switch) were connected in this way using heat shrink. This allowed components to be unplugged and swapped out more easily.
	**Processes** • Testing circuit function **Tools/Equipment/Software** • 12V power supply **Components involved** • Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch **Materials involved** • Jumper wires with	This image shows all of the components of the device wired together for testing, as per the circuit diagram (Week 1, Day 1). The battery has been replaced by a 12V battery, as it has not arrived yet. At this stage, the code was not complete, so I used a simplified script. The script ignored any inputs from the

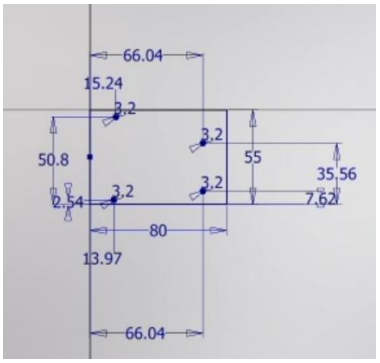
	dupont connectors ● Jumper wires with alligator clips **OH&S considerations** ● Risk Assessment for “12V Power Supply”	rocker switch, and instead had the linear actuator fully extend and retract in a cycle. This confirmed that the linear actuator, arduino, and motor controller were all functional. To test that the rocker switch was functional, I used a program which switched on the built in LED on the Arduino board when the rocker switch was in either the extend or retract position, and turn the LED off when it was in the centre position. This confirmed that the rocker switch functioned correctly.

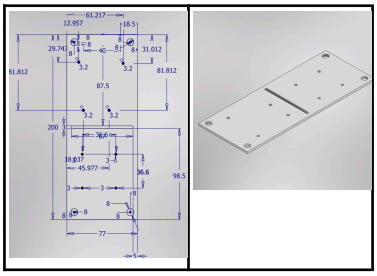
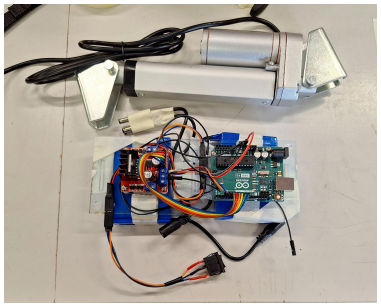
Notes and next steps:

After the successful testing of all of the parts for the device, the following things need to be completed:

- Arduino code which allows the extension and retraction of the linear actuator to be controlled using the rocker switch. (this change is addressed on Week 6, Day 1)

Day 3:

Image	Outline	Explanation
	**Processes** ● CAD Design **Tools/Equipment/Software** ● Autodesk Inventor	The image on the right shows an sketch of a mounting bracket for the Arduino, using the screw hole dimensions and placements specified by the manufacturer. This could allow an enclosure for the arduino and motor controller to be made using laser cut materials available at the school such as 3mm thick acrylic or MDF.
	Processes ● CAD Design ● Laser cutting **Tools/Equipment/Software** ● Autodesk Inventor	The image shows the mounting plate for both the Arduino Uno R3 and the L298N motor controller. The following are features of the design.

	• 40W laser cutter **Components involved** • Arduino Uno R3 • L298N Motor Controller **Materials involved** • 3mm acrylic sheet **OH&S considerations** • Risk Assessment for “Laser Cutter”	• M8 holes to allow for mounting to an enclosure. • M3 holes for mounting the L298N and Arduino to the plate. • Central divider slot 3mm thick, allowing a perpendicular separator to be slotted in the centre if required in the future.
	**Processes** • Laser cutting • Assembly of circuit **Tools/Equipment/Software** • Autodesk Inventor • 40W laser cutter **Components involved** • Arduino Uno R3 • L298N Motor Controller • 2x 12V Lithium ion batteries • Laser-cut acrylic mounting plate **Materials involved** • Masking tape • Hot glue • Jumper wires with dupont connectors **OH&S considerations** • Risk Assessment for “Laser Cutter” • Risk Assessment for “Hot Glue Gun” • Risk Assessment for “Lithium-ion Battery”	This image shows the circuit, as per the wiring diagram, with both the L298N and Arduino mounted using M3 screws to the mounting plate. The batteries have also arrived, and were temporarily mounted to the mounting plate using masking tape before exploring a better solution.

Notes and next steps:

After putting the circuit together in this way, I realised that the following problems need to be addressed:

- The method of using an acrylic plate for the wiring is not suitable, as the acrylic is too flexible for the heavy batteries. (this change is addressed on Week 7, Day 3)
- A way to connect the batteries to the Arduino must be determined, as they use a 2.1mm barrel jack connector. Each individual battery has the same connector, but the issue is that both batteries must be connected in series. So there needs to be a way

to connect the batteries in series, but in a way where each battery is still easily swappable and not soldered together.

- The mounting solution for the wiring must have a sturdy enclosure which fully encloses all of the wiring. After the testing with an open enclosure, it was evident that the wires could easily come loose since they were unprotected. (this change is addressed on Week 7, Day 3)

Week 2 and Week 3:

No progress was made during these weeks due to illness

Week 4

Day 1:

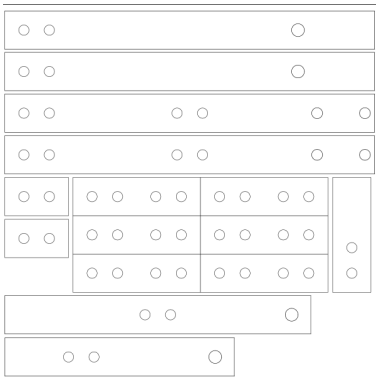
Image	Outline	Explanation
N/A	**Processes** • Test fitting **Tools/Equipment/Software** • Tape measure **Components involved** • 25mm ratchet straps	Today I purchased some ratchet straps to attach the device to the arm of the user. These ratchet straps are 25mm in width and are 1 metre long. There are four straps total, two for the lower arm and two for the upper arm. I tested how the straps fit my arms, and decided that some excess length will need to be cut in the future.

Notes and next steps:

After purchasing and testing the ratchet straps, I made the following observations:

- The ratchet straps are difficult to tighten and secure when they are on your own arm. A better solution could include velcro straps or a buckle, so I later decided to purchase velcro straps to replace the ratchet straps.

Day 2:

Image	Outline	Explanation
	**Processes** • Water-jet cutting • Verification of dimensions for parts **Tools/Equipment/Software** • Adobe Illustrator • Water-jet cutter • Digital callipers **Materials involved** • 300x300x8mm aluminium sheet **OH&S considerations** • Risk Assessment for “Water-jet Cutter” • Risk Assessment for “Digital Calipers”	Today, the metal parts for the device were made by using a water-jet cutter. A single piece of aluminium with dimensions 300x300x8mm was used to cut all the parts out of. After cutting the parts, they were inspected for defects or any rough edges, and then we tested all the screw holes for a correct fit. All of the parts had dimensions within the tolerances, and no parts needed to be remade.

Day 3:

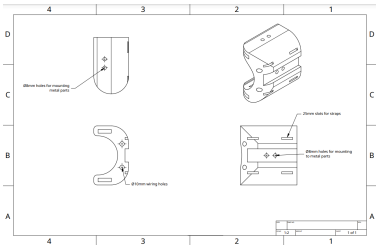
Image	Outline	Explanation
	Processes ● Testing of battery ● Disassembly of battery housing **Tools/Equipment/Software** ● Multimeter **Components involved** ● 2x 12V Lithium ion batteries **OH&S considerations** ● Risk Assessment for “Lithium-ion Battery” ● Risk Assessment for “Hot Glue Gun”	Today, the 12V batteries failed. After using a multimeter to test the voltage, the output read 4V, indicating that a short circuit had blown a fuse within the battery cells. Upon removing the battery cover it was determined that the battery management system for the cells had burned out after a short circuit. I began to search for an alternative battery due to this failure.

Notes and next steps:

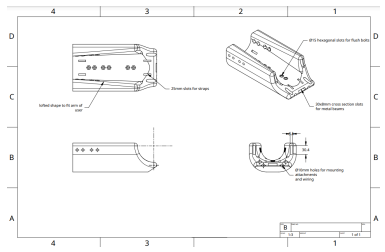
- This failure of the battery led to more caution when using batteries. In the future, I will ensure to turn the battery off before moving it or handling it.
- A new alternative to power the device will need to be explored, as these batteries are no longer viable.

Week 5:

Day 1:

Image	Outline	Explanation
	**Processes** ● CAD Design **Tools/Equipment/Software** ● Autodesk Inventor ● OnShape	This image shows the completed 3D design of the plastic 3D printed part for the upper arm (one of two plastic structural parts in the device). The key features in the design of this part include: Bottom left: ● 3 rectangular holes with a 30x8mm profile, for mounting the 3 aluminium bars which run parallel to the upper arm. ● 2 circular holes with 10mm diameter for any possible wiring which may need to be routed through. Top left: ● M8 screw holes for securing the aluminium bars to the plastic part. ● Circular fillet with large diameter to minimise any possible contact with the upper part of the assembly, and to save material. Top right: ● More M8 screw holes for securing aluminium bars to plastic part Bottom right:

		• Slots for straps to be inserted with 25mm width. Not visible: • Hexagonal indents for the bolt holes on one side, so that the hex nuts can be inserted flush with the walls. **Note:** All slot dimensions and hole dimensions were increased in post by 0.2mm to allow tolerance for fit after 3D printing.

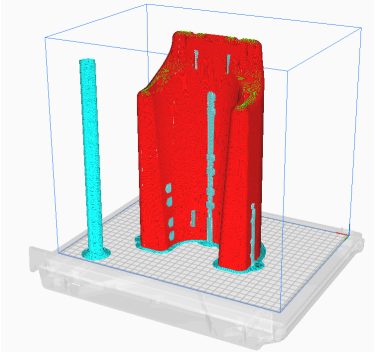
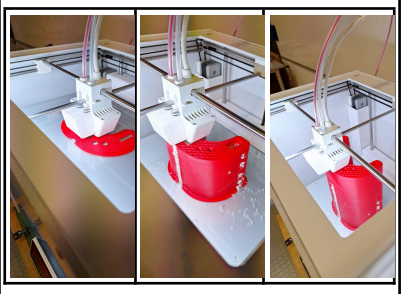
	**Processes** ● CAD Design **Tools/Equipment/Software** ● Autodesk Inventor ● OnShape	This image shows the completed 3D design of the plastic 3D printed part for the lower arm. The key features in the design of this part include: Bottom left: ● M8 screw holes for securing the aluminium bars to the plastic part. ● Circular fillet with large diameter to minimise any possible contact with the upper part of the assembly, and to save material. Top left: ● Hexagonal indents for the bolt holes on one side, so that the hex nuts can be inserted flush with the walls. ● Lofted inside wall allows the inside surface to fit the shape of the human arm. ● Slots for straps to be inserted with 25mm width. Top right: ● More M8 screw holes for securing aluminium bars to plastic part Bottom right: ● 3 rectangular holes with a 30x8mm profile, for mounting the 3 aluminium bars which run parallel to the upper arm.

		• 2 circular holes with 10mm diameter for any possible wiring which may need to be routed through. **Note:** All slot dimensions and hole dimensions were increased in post by 0.2mm to allow tolerance for fit after 3D printing.

Notes and next steps:

- Since the models to be 3D printed are very large, I need to start 3D printing them as soon as possible as it will take a significant amount of time.

Day 2:

Image	Outline	Explanation
	**Processes** • Slicing of STL files **Tools/Equipment/Software** • Ultimaker Cura	Today I started 3D printing the main plastic structural part for the lower arm, which holds the mechanism together. I used PLA as the main material, and PVA supports. The PVA supports were chosen so they can dissolve when submerged in water, allowing effective removal of supports. Total print duration is 5 days, 6 hours. Total mass of the part is 0.958kg.
	**Processes** • 3D printing **Tools/Equipment/Software** • Ultimaker S5 3D Printer **Materials involved** • PLA filament • PVA support filament **OH&S considerations** • Risk Assessment for “3D Printer”	Photos of the 3D printing process of the main lower arm piece.

Notes and next steps:

- Since the other 3D printed part will also take a substantial amount of time to print, this needs to be commenced as soon as possible, and will have to use a different printer. This means that no dissolvable supports can be used with it, as the available printers do not support this.

Day 3:

Image	Outline	Explanation
	Processes • Assembly of tightly fitting parts **Tools/Equipment/Software** • Mallet • Pliers • 40 grit sandpaper **Components involved** • Lower arm cast • Aluminium beams for lower arm • 12x M8 hex bolts and nuts **OH&S considerations** • Risk Assessment for "Mallet" • Risk Assessment for "Pliers" • Risk Assessment for "Sandpaper"	The image on the left shows the finished plastic cast piece for the lower arm. It also shows the M8 hex bolts used to secure the metal beams which support the structure of the plastic part. The image shows some of the metal beam parts already inserted within their respective slots. The metal parts had a tight fit, and required a mallet to fully insert. One of the metal parts was slightly too thick at one end, so it was sanded to create a wedge shape so it could fit in the slot.
	Processes • Test fitting **Components involved** • Lower arm assembly	This image shows me testing the fit of the lower arm cast component. I have inserted the 25mm straps through the slots built into the model, and have tightened it to fit my arm.

Notes and next steps:

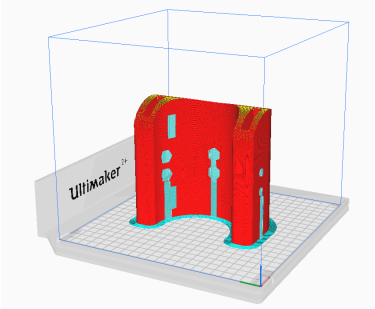
- As the lower arm piece was printed without any issues, no adjustments need to be made for the upper arm piece, and it can be printed ASAP.
- One of the metal parts was slightly too thick to fit at one end, so it was sanded to a wedge shape.

Week 6:

Day 1:

Code	Outline	Explanation

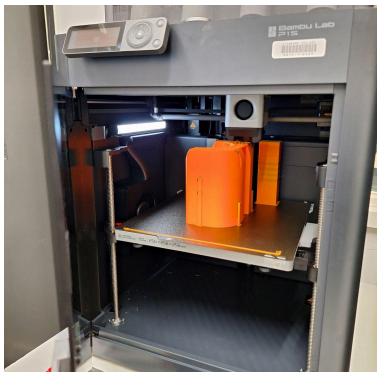
``
#define enB 7
#define in4 6
#define in3 5
#define button1 4
#define button2 3
 	
void setup() {
  pinMode(enB, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(button1, INPUT_PULLUP);
  pinMode(button2, INPUT_PULLUP);
 	
  // Stop the motor initially
  digitalWrite(enB, LOW);
}
 	
void loop() {
  // Read the states of pin 3 and
pin 4
  bool button1State =
digitalRead(button1);
  bool button2State =
digitalRead(button2);
 	
  if (button1State == LOW) {
    // If switch ON pin 3, retract
the actuator
    digitalWrite(in4, LOW);
    digitalWrite(in3, HIGH);
    digitalWrite(enB, HIGH);
  } else if (button2State == LOW) {
    // If switch ON pin 4, extend
the actuator
    digitalWrite(in4, HIGH);
    digitalWrite(in3, LOW);
    digitalWrite(enB, HIGH);
  } else {
    // If switch OFF, stop
    digitalWrite(enB, LOW);
  }
}
``
	Processes
	• Arduino programming
	• Testing of code
	Tools/Equipment/Software
	• Arduino IDE
	• 12V power supply
	Components involved
	• Arduino Uno R3
	• L298N motor controller
	• Linear actuator
	• 3 position rocker switch
	Materials involved
	• Jumper wires with dupont connectors
	• Jumper wires with alligator clips
	OH&S considerations
	• Risk Assessment for “12V Power Supply”
	• Risk Assessment for “Linear Actuator”
		Today I wrote an arduino program for the device, and tested it with the device. Since the batteries failed, I used a 12V power supply in it’s place.
		
		The current iteration of the program is shown on the left. It sends commands to the L298N motor controller to extend or retract the linear actuator based on the current input of the SPDT rocker switch.

Image	Outline	Explanation
	**Processes** • Slicing of STL files **Tools/Equipment/Software** • Ultimaker Cura	Additionally, I started 3D printing the cast for the upper arm, which holds the mechanism together. I used PLA as the main material, with PLA breakaway supports. Total print duration is 23 hours. Total mass of the part is 0.233kg.

Notes and next steps:

- The current code does not function at all, as the actuator does not extend or retract correctly. Further troubleshooting will be needed to determine the cause. (this change is addressed on Week 7, Day 1)

Day 2:

Image	Outline	Explanation
N/A	**Processes** • Testing circuit function **Tools/Equipment/Software** • 12V power supply • Multimeter **Components involved** • Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch **Materials involved** • Jumper wires with dupont connectors • Jumper wires with alligator clips **OH&S considerations** • Risk Assessment for **“**12V Power Supply” • Risk Assessment for “Multimeter”	Today I tried to run the wire the circuit as per the circuit diagram, and then test the circuit. I used a multimeter to test the voltage output of the L298N depending on the physical input of the rocker switch. As a result of testing, one of the Arduinos failed due to a wire being placed in the 5V pin instead of the 12V pin.
	**Processes** • 3D printing **Tools/Equipment/Software** • Ultimaker S5 3D Printer **Materials involved** • PLA filament **OH&S considerations** • Risk Assessment for “3D Printer”	This image shows the upper arm piece being 3D printed.
	Processes • Test fitting **Components involved** • Upper arm cast • 25mm ratchet straps	This image (right) shows the straps attached to the upper arm piece. I also tested the fit of the

		upper arm cast piece on my arm (left).

Notes and next steps:

- Another arduino will need to be purchased for the device to replace the damaged one.
- I must ensure that I carefully follow the wiring diagram in the future before connecting the arduino to DC power. I will take more care in the future to avoid further issues with this.

Day 3:

No progress was made today.

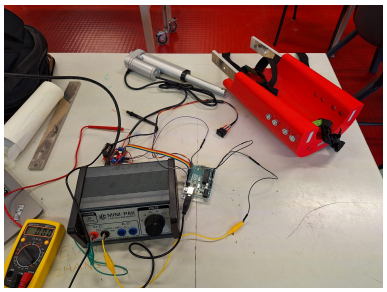
Week 7:

Day 1:

Code	Outline	Explanation

``
// Motor B connections
int enB = 10;
int in3 = 9;
int in4 = 8;
 	
// switch connections
int sw1 = 3;
int sw2 = 4;
 	
void setup() {
  // Set all the motor control pins to outputs
  pinMode(enB, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(sw1, INPUT_PULLUP);
  pinMode(sw2, INPUT_PULLUP);
 	
  // Turn off motors - Initial state
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}
 	
void loop() {
  int sw1_state = digitalRead(sw1);
  int sw2_state = digitalRead(sw2);
 	
  if (sw1_state == LOW) {
    // If switch ON pin 3, retract the actuator
    analogWrite(enB, 255);
    digitalWrite(in3, HIGH);
    digitalWrite(in4, LOW);
  } else if (sw2_state == LOW) {
    // If switch ON pin 4, extend the actuator
    analogWrite(enB, 255);
    digitalWrite(in3, LOW);
    digitalWrite(in4, HIGH);
  } else {
    // If switch OFF, stop
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
  }
}
``
	Processes
	• Testing circuit function
	• Arduino programming
	Tools/Equipment/Software
	• 12V power supply
	• Multimeter
	• Arduino IDE
	Components involved
	• Arduino Uno R3
	• L298N Motor Controller
	• 12V Linear Actuator
	• 3 position rocker switch
	Materials involved
	• Jumper wires with dupont connectors
	• Jumper wires with alligator clips
	OH&S considerations
	• Risk Assessment for “12V Power Supply”
	• Risk Assessment for “Multimeter”
		Using this code, the linear actuator now extends and retracts according to the position of the rocker switch. The following issues were resolved to achieve this:
		• Faulty arduino board
		• Incorrect wiring of switch
		• Incorrect wiring of actuator

Day 2:

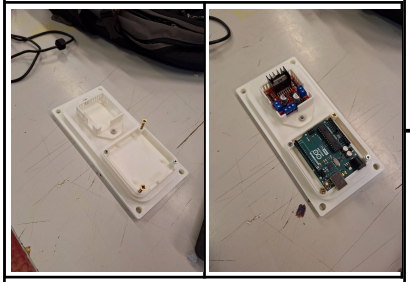
Image	Outline	Explanation
	**Processes** • Testing circuit function • Arduino programming **Tools/Equipment/Software** • 12V power supply • Multimeter • Arduino IDE **Components involved**	Today I continued to work on the code for the arduino. I determined that there was an error with the power supply I was using to test it, which led to a higher voltage than normal (18V). I used a multimeter to test it, and tuned it to the correct

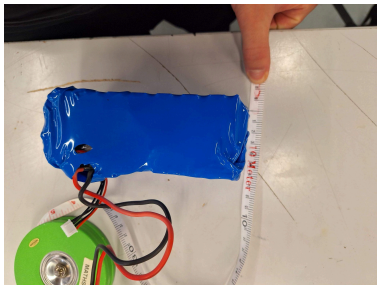
	• Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch **Materials involved** • Jumper wires with dupont connectors • Jumper wires with alligator clips **OH&S considerations** • Risk Assessment for “12V Power Supply” • Risk Assessment for “Multimeter”	12V voltage. The modified code is shown below:

Notes and next steps:

- The next thing to do is to assemble the cast for the upper arm, and decide where to mount the battery.
- Also, a housing for the toggle switch needs to be created, which can ergonomically fit into the palm of the hand of the user. (this change is addressed on Week 8, Day 1)

Day 3:

Image	Outline	Explanation
	**Processes** • 3D printing • Slicing of STL files • CAD Design **Tools/Equipment/Software** • Autodesk Inventor • Bambu Slicer software • Bambu X1 Carbon 3D Printer **Components involved** • Arduino Uno R3 • L298N motor controller **Materials involved** • PLA filament **OH&S considerations** • Risk Assessment for “3D Printer”	Today I 3D printed enclosures for the L298N, the Arduino Uno R3, and a project box to contain both of those. I also printed a controller for the device, which has a rocker switch which press-fits into a cylinder.

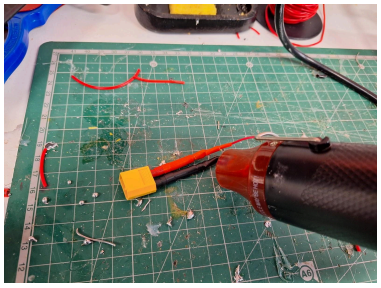
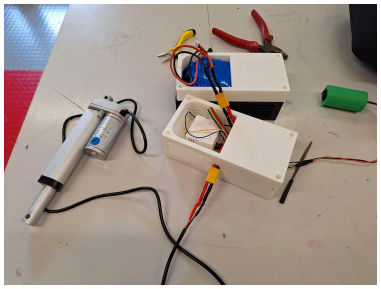
	**Processes** • Measuring of components **Tools/Equipment/Software** • Tape measure **Components involved** • Large 12V battery **OH&S considerations** • Risk Assessment for “Lithium-ion Battery	This image shows the use of a tape measure to take the measurements of the new battery.

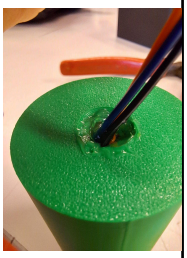
Notes and next steps:

- With the dimensions of the battery taken, the next step is to design a 3D printed enclosure for the battery.

Week 8:

Day 1:

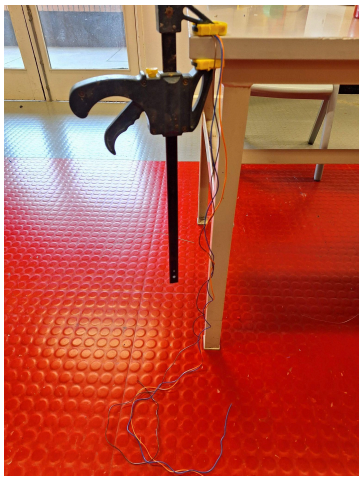
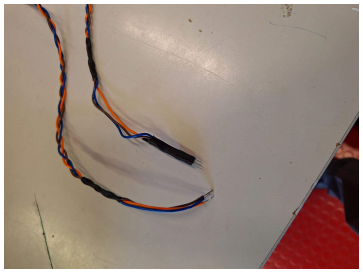
Image	Outline	Explanation
	**Processes** • Soldering • Heat shrinking **Tools/Equipment/Software** • Heat gun • Soldering iron **Components involved** • XT60 connectors **Materials involved** • 3mm heat shrink **OH&S considerations** • Risk Assessment for “Heat Gun” • Risk Assessment for “Soldering Iron”	Using a heat gun to heat shrink solder connections. In the image I soldered longer wires to the XT60 connector for the L298N, so that I could easily connect and disconnect the battery from the L298N.
	**Processes** • Assembly of circuit **Components involved** • Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch • Large 12V battery • 3x circuit box components • 1x ergonomic housing for rocker switch **Materials involved** • Jumper wires with dupont connectors **OH&S considerations** • Risk Assessment for “Lithium-ion Battery	Today I assembled the main circuit box, and wired everything together as per the circuit diagram. The goal for today was to wire everything together so that the circuit functions correctly while completely enclosed.
	Processes • Securing wires with hot glue **Tools/Equipment/Software** • Hot glue gun **Components involved**	I used a hot glue gun to reduce strain on any areas where wires should be permanently slotted through holes, effectively functioning similarly to a strain relief.

	• XT60 connectors **Materials involved** • Hot glue sticks **OH&S considerations** • Risk Assessment for “Hot Glue Gun”	The green plastic part is an ergonomic housing for the toggle switch for the control of the device.

Notes and next steps:

- Circuit box internals are completed. Next step is to connect all parts of the assembly and test the entire device as a functioning system.

Day 2:

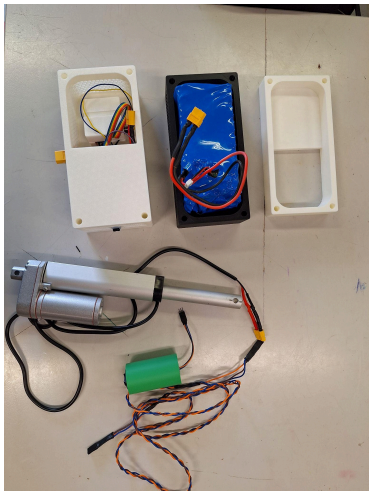
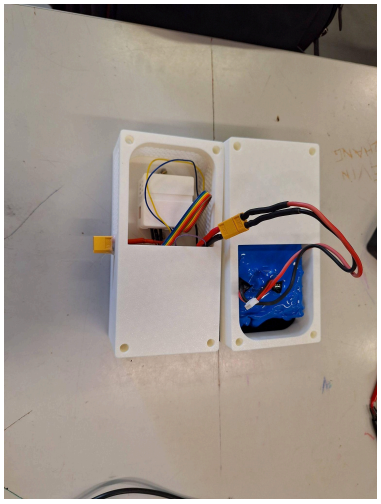
Image	Outline	Explanation
 	**Processes** • Cable braiding **Tools/Equipment/Software** • Clamp **Materials involved** • 3x 2m long wires	Today I worked on creating a long braided cable which has dupont connectors on both sides. This allows a connection between the control switch and the arduino which is stored in the circuit box. The cable is about 1.5 metres long, and includes 3 wires. (ground wire is central)

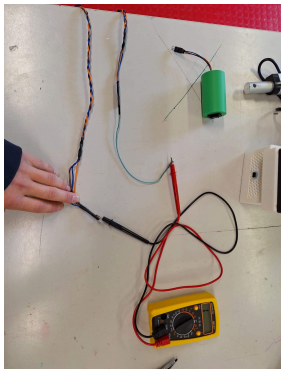
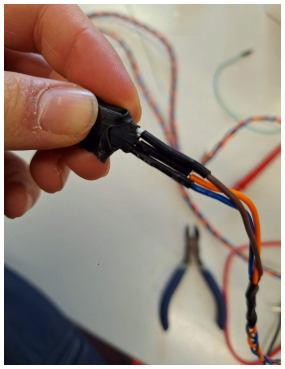
Notes and next steps:

- Same as previous day notes.

Week 9:

Day 1:

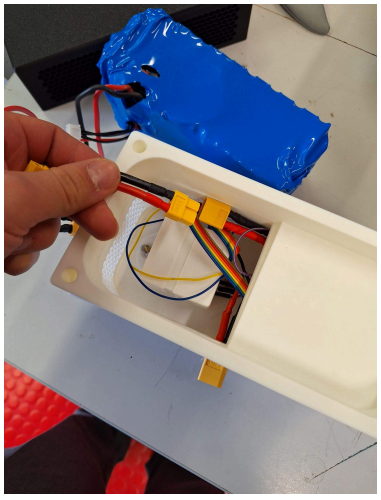
Image	Outline	Explanation
	**Components involved** • Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch • Large 12V battery • 3x circuit box components • 1x ergonomic housing for rocker switch **OH&S considerations** • Risk Assessment for “Lithium-ion Battery	An image showing all the different electronic components as well as the plastic parts for the circuit box.
 	**Components involved** • Arduino Uno R3 • L298N Motor Controller • 12V Linear Actuator • 3 position rocker switch • Large 12V battery • 3x circuit box components • 1x ergonomic housing for rocker switch **OH&S considerations** • Risk Assessment for “Lithium-ion Battery	This image shows the battery connected to the main circuit (left). The right image shows the full functioning circuit box with the lid secured on top.

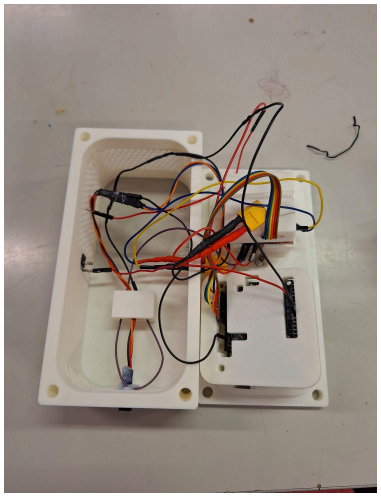
 	**Processes** • Continuity testing **Tools/Equipment/Software** • Multimeter **Components involved** • Long braided 3-wire cable **OH&S considerations** • Risk Assessment for “Multimeter”	After testing the circuit using the long braided cable, the switch was not functioning correctly. After performing a continuity test with a multimeter (left), it was determined that the wires in the long cable connector were in the wrong order (right). The correct order has ground in the middle so that the connector can be two directional. Once this was resolved, the circuit functioned as intended.

Notes and next steps:

- Proceed to testing as soon as possible.

Day 2:

Image	Outline	Explanation
	**Processes** • Soldering • Heat shrinking **Tools/Equipment/Software** • Soldering iron • Heat gun **Components involved** • XT-60 connectors **Materials involved** • Spare wire segments **OH&S considerations** • Risk Assessment for “Soldering iron” • Risk Assessment for “Heat gun”	After attempting to disconnect the battery to prevent discharge (left), the cable attaching the battery snapped due to the high force required to pull apart the connectors. The assembly was disassembled and the broken cables were re-soldered.

		
	Processes ● Soldering ● Heat shrinking ● Modification of plastic component **Tools/Equipment/Software** ● Soldering iron ● Heat gun ● Cordless drill **Components involved** ● Circuit box assembly ● Rocker toggle switch **Materials involved** ● Spare wire segments **OH&S considerations** ● Risk Assessment for “Soldering Iron” ● Risk Assessment for “Heat Gun” ● Risk Assessment for “Cordless Drill”	After the incident with the battery, I decided to drill a hole into the side and file it to create a square slot for a battery toggle switch, which I wired directly to the battery circuit. This allows the battery to be toggled to prevent discharge.

Notes and next steps:

- All circuitry and mechanical components are complete. The system can now be assembled and tested.

Day 3:

Image	Outline	Explanation
 	Processes ● Greasing of joints **Components involved** ● Circuit box assembly ● Rocker toggle switch **Materials involved** ● White lithium grease **OH&S considerations** ● Gloves used to avoid skin contact with grease	Application of lithium grease to load bearing joints. Gloves used to avoid skin contact with potentially hazardous chemicals.

Criteria 6: Diagnostic Testing

Test 1: Range of Motion

Aim	To measure the range of motion of the device.
Expected Result	As calculated in Criteria 2, the range of motion should be approximately 60° in theory.

Equipment

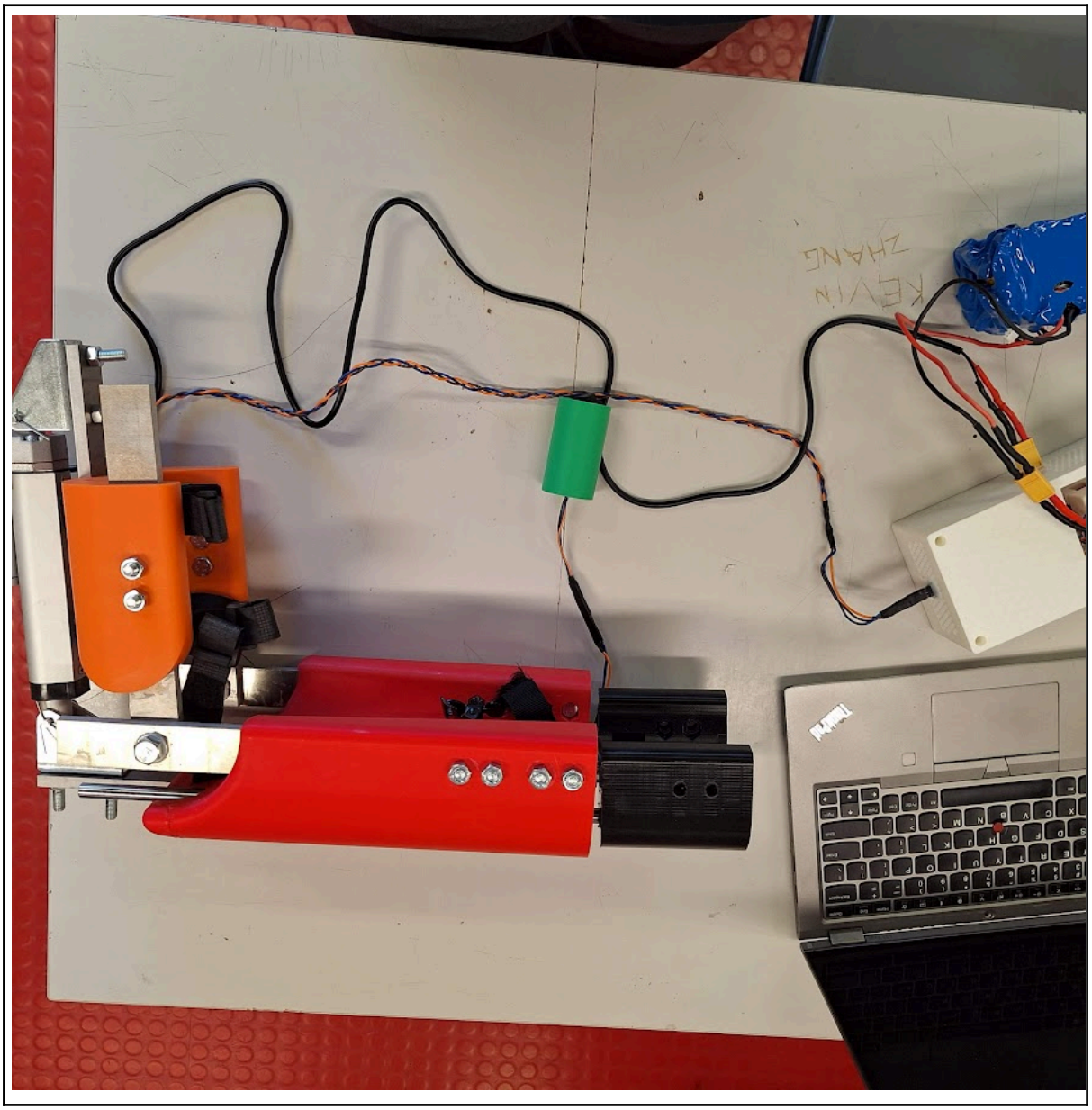
- ExoArm device
- Protractor

Method

1. Set the ExoArm device to a fully contracted position
2. Use protractor to measure the angle between forearm and upper arm sections, and record this first angle
3. Set the ExoArm device to a fully extended position
4. Use protractor to measure the angle between forearm and upper arm sections, and record this second angle
5. Subtract the first angle measured from the second angle to obtain the range of motion

Results

The angle measured from the range of motion test was approximately 60 degrees.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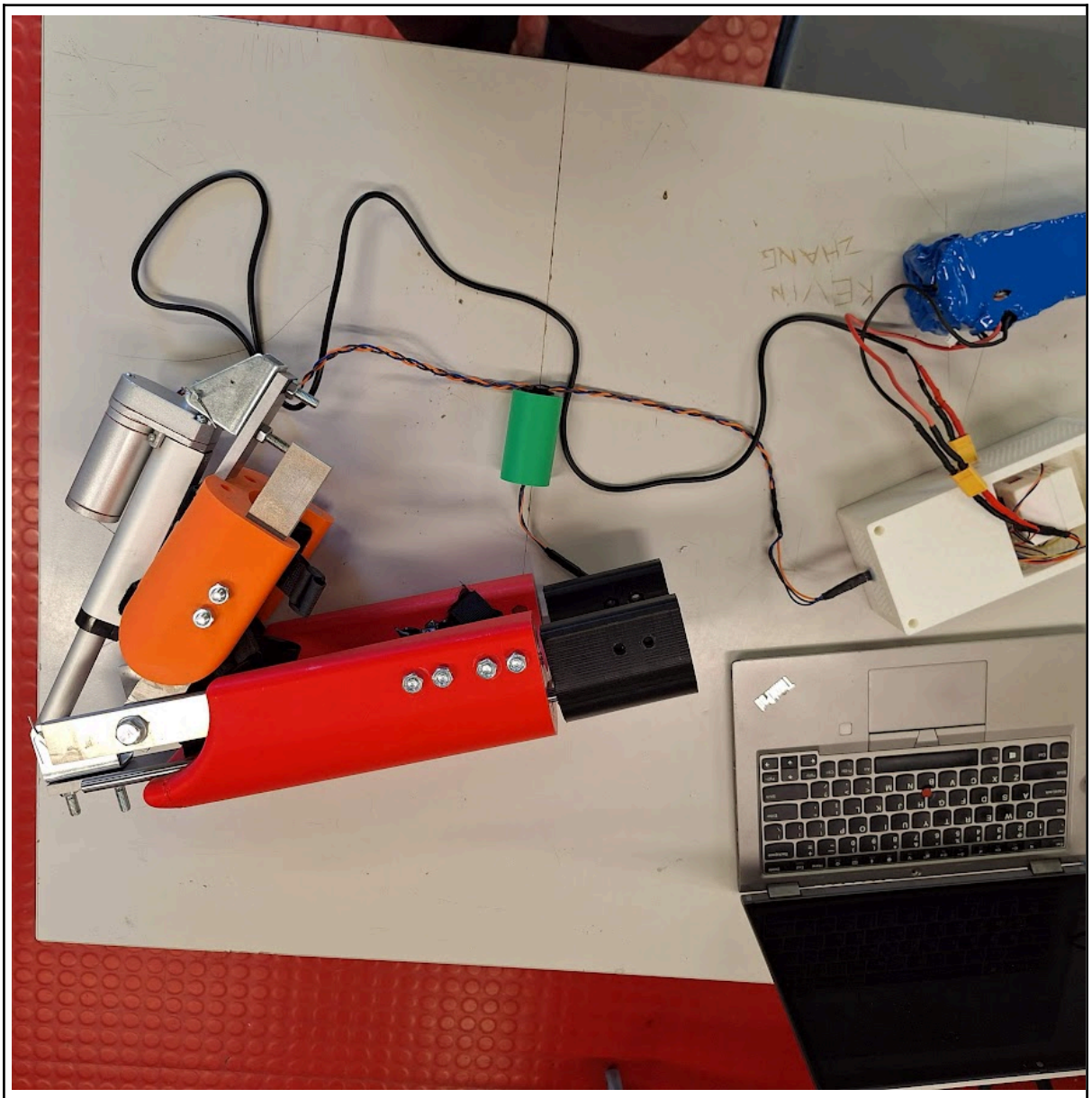


Discussion

These results align with the expected results for the device.

Test 2: Force Output

Aim	To test the the effectiveness of the device in assisting with lifting a load
Expected Result	The device should enable user to lift more load that the user can normally hold OR lift the same load with lower difficulty.

Equipment

- ExoArm device
- Human tester
- Assorted dumbbells

Method

1. Human tester to bicep curl a dumbbell with they can lift for 1 repetition with a Rate of Perceived Exertion (RPE) of 7
2. Human tester then equips the ExoArm device, ensuring that all straps are securely fastened.
3. Human tester to bicep curl the same weight and record the RPE.
    a. IF the RPE is below 7, increase the weight until an RPE of 7 is reached, and then record that weight.

Results

Discussion

Test 3: Load-bearing capacity

Aim	To test the durability of the device under load
Expected Result	Under the loads being tested in **Test 2**, there should be no cracking or deformation

Equipment

- ExoArm device
- Human tester
- Assorted dumbbells

Method

1. After the completion of **Test 2**, inspect the device for any cracks or deformation.

Results

Discussion

Test 4: Latency

Aim	To test the time delay between the input to the device and the response by the device.
Expected Result	The latency should not be noticeable (500ms or less)

Equipment

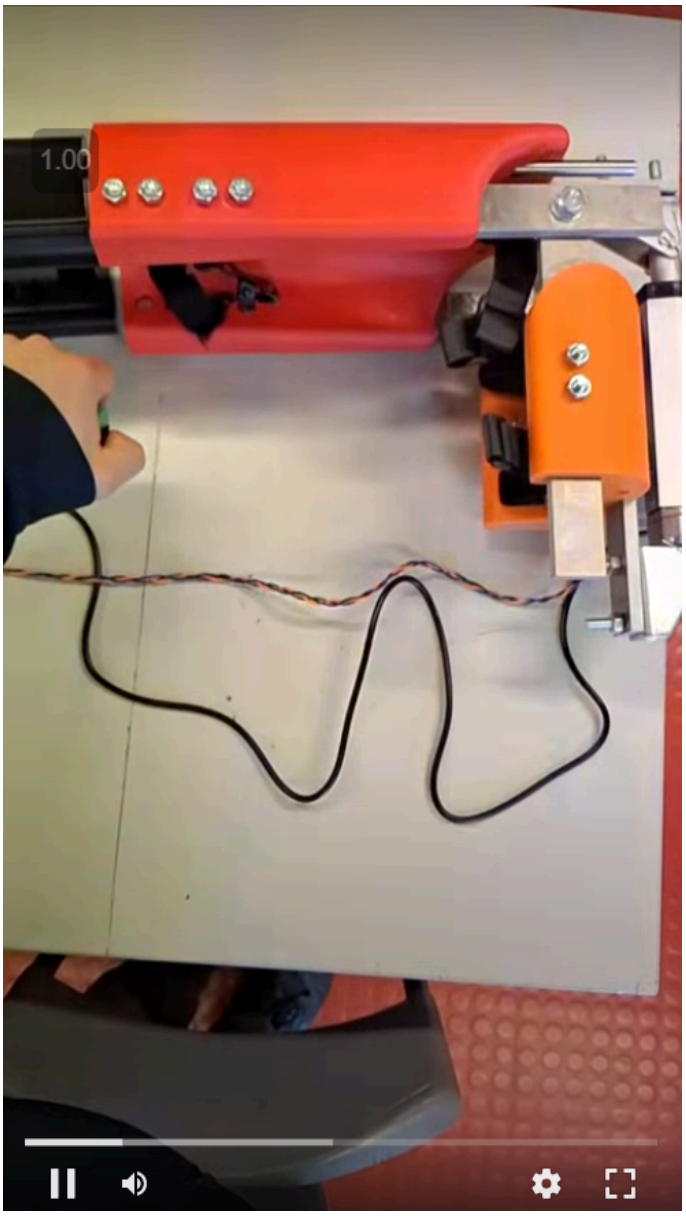
- ExoArm device
- Video camera

Method

1. Set the ExoArm control switch to a neutral (centre) position
2. Start recording a video which includes both the ExoArm device and the control switch in the frame.
3. Set the control switch to extension.
4. From the recording, find the time delay between when the switch is toggled and the actuator extends.

Results

After the switch was toggled, the time delay was approximately 1/8th of a second, or 125ms.



Slow motion recording of the toggling on and off of the device, which allowed latency to be determined

Discussion

These results exceeded the expected results of the device.

Test 5: Comfort and Ergonomics

Aim	To test the comfort of the device when being used.
Expected Result	Users will likely have various points of feedback to improve comfort.

Equipment

- ExoArm device
- Human tester

Method

1. After **Test 2**, ask the user if there are any points of discomfort.

Results

The main points of discomfort from user feedback included:

1. Tightness of velcro straps creating pressure points
2. Weight of the device
3. Range of motion of the device is too small
4. Lack of padding for the device

Discussion

This aligns with the expected results for the device.

Test 6: Speed Test

Aim	To test the time taken between full extension and retraction
Expected Result	The time taken should be approximately 15 seconds.

Equipment

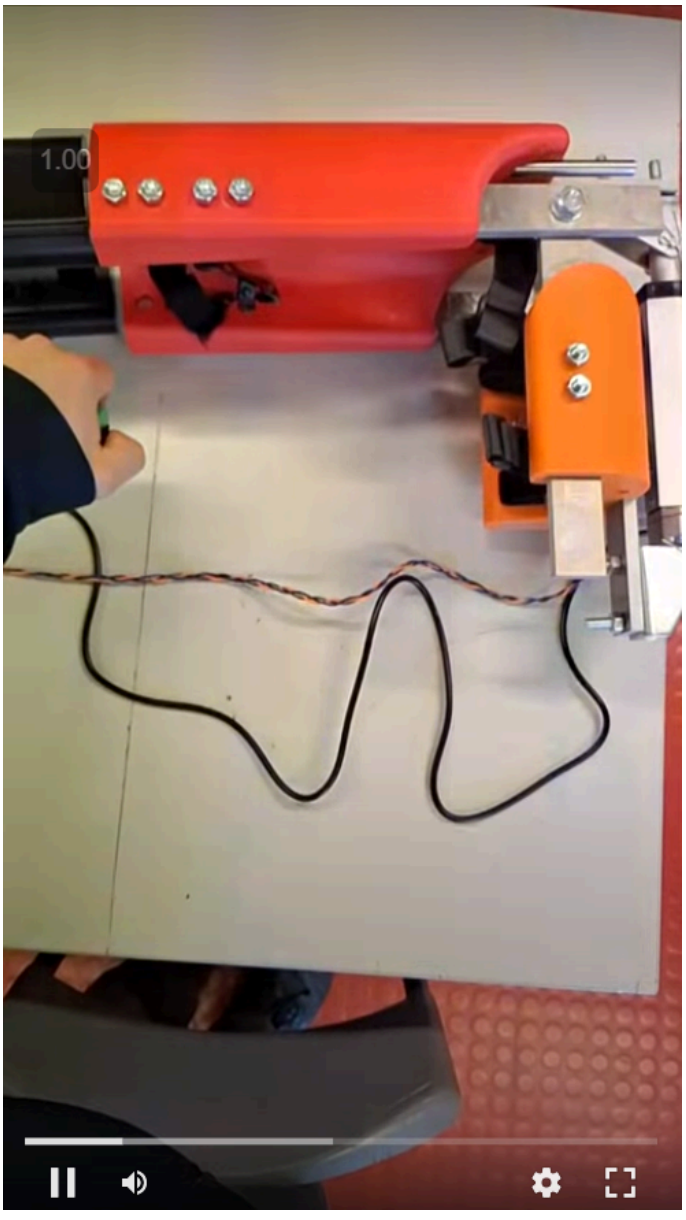
- ExoArm device
- Stopwatch

Method

1. Set the ExoArm device to a fully contracted position
2. Start stopwatch, and simultaneously set the toggle switch to extension
3. Stop stopwatch when the ExoArm device is in a fully extended position

Results

After the switch was toggled, the time taken was 18 seconds for a full extension.



Video recording of the toggling on and off of the device, which allowed speed to be determined
Discussion
These results were slower than the expected results of the device.

Test 7: Weight Test

Aim	To test the overall weight of the device (excluding battery and circuit)
Expected Result	The device will weigh approximately 3 kilograms.

Equipment

- ExoArm device
- Scale

Method

1. Place ExoArm device on scale and record weight

Results

The weight of the ExoArm device was measured to be 5.1kg, excluding the battery and the device.

Discussion

This was heavier than the expected results for the device, likely due to the density of the 3D prints being higher than expected.

Criteria 8: Evaluation of design, production method, and final system

Evaluation Criteria:

Was the project completed within the $100 budget?

Low (1)	Medium (2)	High (3)
The final cost of the project after completion is above $110	The final cost of the project after completion is between $110 and $90	The final cost of the project after completion is under $90

The total cost of the project included the following:

Component	Cost
Lower arm cast • 0.958kg of PLA material • 0.147kg of PVA support material	$29.70 + $24.97 = $54.67
Upper arm cast • 0.233kg of PLA material	$5.36
Circuit box • 0.877kg of PLA material	$20.07
1500N Linear actuator 12V	$49.39
Switches • Rocker switches for control • Toggle switch for battery	$2.00 + $3.17 = $5.17
Aluminium sheet	$73.28
12V batteries	$53.72
Arduino Uno R3	$19.99
L298N motor controller	$16.95
Total	$298.6

Was the project completed within the time constraint of 20 weeks?

Low (1)	Medium (2)	High (3)
The project was completed in more than 20 weeks	The project was completed in 15-20 weeks	The project was completed in less than 15 weeks

Is the device must be lightweight enough to avoid inducing muscular strain or mechanical strain on joints?

Low (1)	Medium (2)	High (3)
User trials reported significant discomfort pertaining to the weight of the device	User trials reported minor discomfort pertaining to the weight of the device	User trials reported no discomfort pertaining to the weight of the device

Does the device allow freedom of arm movement?

Low (1)	Medium (2)	High (3)
The range of motion is below 60% of the range of motion of the arm joint	The range of motion is between 60-90% of the range of motion of the arm joint	The range of motion is above 90% of the range of motion of the arm joint

How much does the device increase the effective pulling and pushing force of the arm?

Low (1)	Medium (2)	High (3)
The device does not increase the amount of weight able to be lifted	The device increases the amount of weight that can be lifted by a measurable amount	The device increases the amount of weight that can be lifted by 50%

Are the tolerance range and speed limits effective, avoiding excessive ranges of motion or speeds?

Low (1)	Medium (2)	High (3)
User trials reported that the	User trials reported that the	User trials reported no

device exceeded the range of motion limit of their arm OR user trials reported that the speed of movement was too fast to be controllable	device approached the range of motion limit of their arm OR user trials reported that the speed of movement was higher than desired	issues with range of motion limits or overspeeding issues

Is the device simple to operate, adjust, and disengage?

Low (1)	**Medium (2)**	**High (3)**
The time taken to put the device on and activate it is more than 5 minutes	The time taken to put the device on and activate it is between 3 and 5 minutes	The time taken to put the device on and activate it is less than 3 minutes

Is the device easily repairable?

Low (1)	**Medium (2)**	**High (3)**
All components rely on soldered connections and are are permanently secured in place.	Most components have standard connectors, however some parts are less accessible and/or glued in place	All components have standard connectors and are easily accessible and replaceable

Is the input latency of the device low enough to be able to use it effectively?

Low (1)	**Medium (2)**	**High (3)**
The input latency is above 500ms	The input latency is between 150ms and 500ms	The input latency is lower than the average human reaction time (>150ms)

Are the structural components of the system strong enough to bear heavy loads?

Low (1)	**Medium (2)**	**High (3)**
The system has major cracks or deformations under the loads tested	The system has minor cracks or deformations under the loads tested	The system system does not crack or deform under the loads tested

Overall Evaluation:

The total score for the system is **19/30** or 63.3%.

During the planning of the project, the overarching details of the device, such as the type of lever being used, and the means of transferring force, were extensively considered and mathematically modelled. This included information such as speed ratios, mechanical advantage, and animations and diagrams. However, some details in the planning stage were neglected due to time restraints, and were finalised during the production of the device. For example, the circuit diagram and outline of the code were both not considered until the early weeks of production, rather than in the planning stage. Additionally, the design of the 3D printed plastic parts, such as the upper and lower arm cast sections, and the circuit component box, were both not finalised until later in production. Due to this, the extensive time and effort using CAD software to design the plastic parts, and the different details and features of the plastic parts, were not discussed deeply in the report.

To improve on this, more time should be allocated for the planning stages. This allows full, labelled 3D models of each prototype, including wiring diagrams and 3D models of plastic parts, to be explained and considered in the planning stage of the report. Additionally, by reducing the amount of constraints and considerations for features of the device, more time and effort can be put into showcasing all the design decisions made in the documentation. This would allow the documentation to be more representative of the time and effort put into the design and production of the device.

The production stages went well compared to the planning stages. Various risk assessments were sourced to minimise any possible risks when using any equipment, and a detailed development roadmap was outlined and followed. Despite getting sick multiple times throughout the development, and losing almost 3 full weeks of time for development and production of the device, the development plan was able to be closely followed with few deviations. The use of water-jet cutting for the production of parts was also very successful. Again, having lower expectations for the features device would have improved this production process even further and improved the quality of documentation.

In terms of sustainability, there is significant room for improvement. The 3D printed plastic parts were very large, and required an extensive amount of time to 3D print, leading to a high risk of failure and plastic wastage. Additionally, due to cost concerns, regular PLA filament was used instead of the planned carbon fibre reinforced nylon filament. To improve on the plastic parts of the device, I believe that the thickness of the parts should be significantly reduced, and the parts should be able to be printed in sections to reduce the amount of very large and risky (over 5 day) 3D prints. Additionally, the use of PVA support material was not effective, as they left residue and did not fully dissolve as much as desired. In the future parts should be designed without PVA support material, to reduce the risk of residue stuck in hard to reach areas (such as the area where the straps are slotted in). Once again, many issues with the design could be resolved by lower expectations. For example, a lower constraint for the maximum load for the device (such as 20kg instead of 60kg) would have allowed the production of the device to be more sustainable by using less plastic, as the structure would not need to be as strong. Additionally, it would reduce power consumption by allowing a lower force linear actuator to be used, improving energy efficiency, sustainability and affordability.

The diagnostic testing and evaluation criteria of the device was successful, and provided insights into how the device performed against the initial constraints of the design brief. The most major flaws in the design which were discovered in these steps included the high cost, and the user discomfort of the device (due to weight and pressure points). In the future, when designing devices such as exoskeletons of prosthetics, the ergonomics and comfort of the device should be carefully considered, as well as the affordability.

Testing revealed that the device was successful in achieving its original goal of enhancing the strength of the arm of the user, however many compromises were made compared to the original design brief and constraints to reduce costs, time, and complexity of the device. These compromises were made despite the extensive amount of time and considerations that went into the planning, development, and production of the device. This indicated that the initial goals for the device's performance, including the considerations and constraints, may have been too ambitious. Despite this, through the solving of problems many valuable lessons were learned in the development of the exoARM device.